

Verification of multiphysics coupling techniques for modeling of molten salt reactors

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ABSTRACT

Crucial to the development of molten salt reactor (MSR) designs is the application of multiphysics codes to model the tightly coupled neutronics and thermal-hydraulics behaviour of the liquid fuel. However, the verification and validation of such codes is not a trivial task, in particular for fast reactor designs, where no experimental data are available. In absence of experimental data, a benchmark was developed by LPSC/CNRS-Grenoble for multiphysics codes dedicated to MSR studies. In this study we present two independent multiphysics approaches and apply them to this benchmark. The first approach utilizes the Serpent2 multiphysics interface, allowing for high fidelity coupling of the finite volume computational fluid dynamics code OpenFOAM and Serpent2. In this approach, Serpent2 serves as the neutronics solver and is coupled to an OpenFOAM based thermal-hydraulics solver and supplemented by a delayed neutron precursors transport solver implemented in OpenFOAM. The main advantage of this coupling approach is that it allows for using a high accuracy Monte-Carlo approach to solve the neutron transport equations. The second approach is a novel approach that utilizes the SEALION framework. The SEALION code employs a specialized thermal hydraulics solver based on OpenFOAM, coupled with a custom-made modified point kinetics neutronics solver, that explicitly accounts for the altered neutron importance due to the transport of delayed neutron precursors. The main advantage of this approach is that it allows for a pre-determination of the temperature feedback effects using Monte-Carlo codes, such as Serpent2. Both approaches are verified against results from the benchmark and the overall agreement between the results demonstrates the validity of both approaches.

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1. Introduction

Molten salt reactors (MSRs) have recently gained renewed interest due to potential advantages compared to conventional reactors based on solid fuels; these include inherent safety features, low pressure operation, large negative feedback coefficients and the possibility for online fuel reprocessing. MSR designs, however, are typically characterized by a low technology readiness level and several challenges including the corrosion of the structural materials, developing a fuel cycle and protection against proliferation.

Among several conceptual MSR designs, the Molten Salt Fast Reactor (MSFR) has been selected as a Generation IV Reactor candidate (Boussier et al., 2012). The MSFR design concept is outlined

in Fig. 1. The design has evolved over the years from having a cylindrical core to a segmented core geometry in the framework of the European SAMOFAR project (SAMOFAR, 2019). In the segmented design approach the core is toroidal, surrounded by a fertile salt blanket and has sixteen fuel inlets and outlets connected to sixteen heat exchangers. In case of an accident the fuel is supposed to drain into an emergency draining tank which is sealed with a freeze plug during normal operation. The molten fluoride salt enters the toroidal core at a nominal temperature of 650 °C, and the design temperature rise within the core is 100 °C at a nominal flow rate of 4.5 m³/s.

The liquid fuel in the core introduces several fundamental differences compared to a solid fueled reactor. In particular, the delayed neutron precursors (DNP) are no longer stationary, but move with the flow, resulting in a decreased effective delayed neutron fraction. Since the liquid fuel is also the primary coolant, turbulence effects have direct impact on the neutronics performance of the reactor. In addition to this and the application of gaseous

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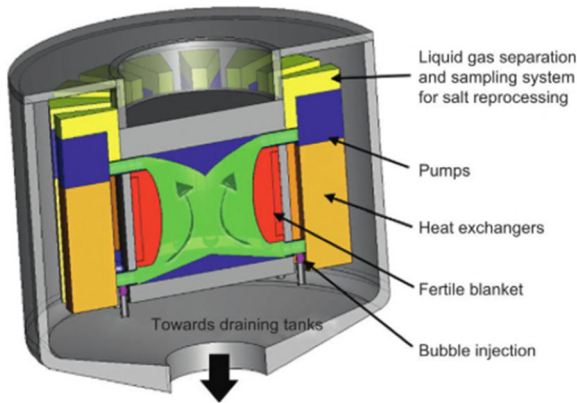


Fig. 1. The outline of the MSFR design.

fission product removal systems, the compressibility of the fuel contribute to the strong coupling between neutron kinetics and thermal hydraulics; thus requiring multiphysics modeling tools to study such reactors. Regulatory approved modelling tools have been developed specifically for solid fuel reactors and even though some of them can be extended or modified to address liquid fuels (Fletcher and Schultz, 1995; Zhang et al., 2018), no multiphysics analysis framework has been approved specifically for liquid fuel reactors. A short summary of some multiphysics tools developed and applied for the analysis of MSRs, in particular the MSFR, is presented in Table 1. A brief description of the codes and further references can be found in (Tiberga et al., 2020).

All codes in Table 1 are based on the solution of the neutron transport equation using either a diffusion or an SP_N approximation (Cervi et al., 2019a; Fiorina et al., 2012; Fiorina et al., 2014). A new contribution to the original fast MSR benchmark developed by LPSC/CNRS-Grenoble (Aufiero, 2015; Laureau, 2015; Aufiero and Rubiolo, 2018) has recently been published (Tiberga et al., 2020). The original benchmark and the recent extension by Tiberga et al. can be used by multiphysics code developers to verify their modeling approaches. The structure of the benchmark, wherein coupling terms are introduced gradually, allows for easy debugging of the codes and identification of the sources for discrepancy. Unless otherwise stated the study here will refer only to the results from the recent publication by Tiberga et al.

For the present benchmark the CNRS code employs either a SP_1 or a SP_3 model for the neutronics combined with a thermal-hydraulics solver based on the finite volume C++ computational fluid dynamics (CFD) toolbox OpenFOAM (Weller et al., 1998). The PoliMi code uses a multi-group diffusion model for the neutronics and OpenFOAM on the thermal-hydraulics side. The PSI code implements a multi-group diffusion sub-solver in the OpenFOAM code. Finally the TUD code employs an S_N multi-group model

for the neutronics and a parallel solver for the incompressible Navier–Stokes equations.

In this paper, we present two multiphysics modeling approaches and verify them against the benchmark. The first approach (Nalbandyan et al., 2019) utilizes the Serpent2 (Leppänen et al., 2015) multiphysics interface (Valtavirta, 2016), allowing for high fidelity coupling of OpenFOAM and Serpent2. In this approach, Serpent2 serves as the main neutronics solver and is coupled to an OpenFOAM based thermal-hydraulics solver and a DNP transport solver implemented in OpenFOAM as well. The main advantage of this coupling approach is that it allows for using a high accuracy Monte-Carlo method to solve the neutron transport equations. In the following this approach will be referred to as the DTU approach. The second approach utilizes the SEALION framework. The SEALION framework, in its current iteration, employs a specialized thermal hydraulics solver based on OpenFOAM coupled with a custom-made modified point kinetics neutronics solver explicitly taking the altered neutron importance from DNP transport into account. The main advantage of this approach is that it allows for pre-determining the temperature feedback using Monte-Carlo codes such as Serpent2, leading to a significant reduction in computational requirements. In general point, kinetics solvers are expected to capture the physics of reactor systems, and provide a good approximation for a given reactor transient, as long as the transient is (mainly) driven by the fundamental form. It is therefore of interest to expand and demonstrate point kinetics methods for liquid fueled reactor systems such as MSRs. Despite their reduced order, point kinetics solvers are still being used extensively in nuclear reactor transient analysis codes such as RELAP5/7 (Fletcher and Schultz, 1995; Zhang et al., 2018).

Both modelling approaches are explained in greater details in Section 4.

In Section 2, a detailed description of the benchmark is provided. Section 3 describes the various phases and steps that constitutes the benchmark. The assumptions for every step of the benchmark as well as the output observables are described in this section. The two code packages used in paper will be presented in Section 4. The results obtained from applying these codes to the benchmark are presented and analyzed in Sections 5 and 6.

2. Description of the benchmark

One of the traditional benchmark cases used to verify thermal-hydraulics solvers for incompressible flow is the lid driven cavity model (Aufiero, 2015; Laureau, 2015). The geometry is fairly simple, it captures the main characteristics of the flow and provides a clear identification of relevant parameters for the simulation. The benchmark geometry has been adapted in (Tiberga et al., 2020) as well as this work for verification of the neutronics-thermal hydraulics coupling techniques for MSRs; the cavity model also represents a simplified Molten Salt Fast Reactor design.

2.0.1. Geometrical layout of the benchmark

The lid driven cavity consists of a 2 m by 2 m domain filled with a molten salt. The benchmark geometry is shown in Fig. 2. As OpenFOAM is unable to deal with purely 2D geometry the cavity is extended along a pseudo z-axis by 1 m. However, no segmenting of the geometry along the z-axis has been used, in order to preserve the 2D aspects of benchmark, reduce complexity and avoid inherent 3D effects like turbulence. The boundary conditions for the walls perpendicular to the pseudo z-axis (front and back faces) are specified as *empty*, which forces the solver to not solve conservation equations for these boundaries. As indicated in Fig. 2, two centerlines AA' and BB' have been introduced to facilitate a point-

Table 1
Multiphysics code used in MSFR benchmark (Tiberga et al., 2020).

Developer	Neutronics code package	Thermal hydraulics code package	Reference
CNRS	Multi-group diffusion and SP_N	OpenFOAM	(Blanco et al., 2020)
PoliMi	Multi-group diffusion and SP_3	OpenFOAM	(Cervi et al., 2019a; Cervi et al., 2019b)
PSI/EPFL ^a	Multi-group diffusion and SP_3	GenFOAM	(Fiorina et al., 2015)
TUD	Phantom S_N	DGFlows	(Tiberga et al., 2019)

^a In the following the PSI/EPFL will be referred to as PSI as has been done in (Tiberga et al., 2020)

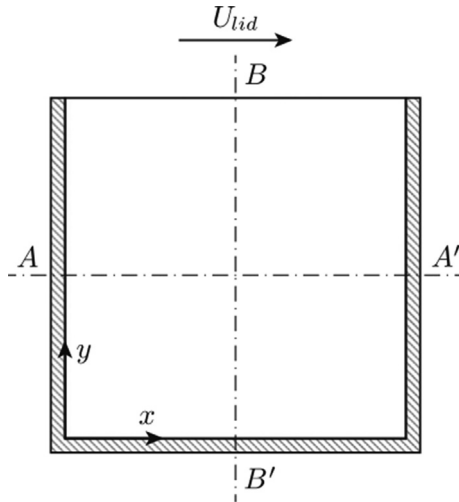


Fig. 2. Benchmark geometry: a simple $2\text{ m} \times 2\text{ m} \times 1\text{ m}$ cavity treated as a 2D problem.

wise comparison of the observables for the various steps of the benchmark.

2.0.2. Input data and boundary conditions: Neutronics

A fluoride salt with U-235 as the fissile element is used in the benchmark, with the salt composition specified in Table 2. While the specific choice of nuclear library for the benchmark is JEFF-3.1 (Koning et al., 2006), the nuclear data used in this paper is taken from JEFF-3.2 (Plompen et al., 2020). The systematic uncertainties introduced by using a different version of the nuclear data library is addressed in Section 5. The benchmark further prescribes that nuclear data is evaluated at the reference temperature $T_{ref} = 900\text{ K}$ only. The motivation for this is that the Doppler broadening effect on microscopic cross sections is negligible; this avoids biases arising from the differences in treatment of this complex effect by each code. As a consequence hereof, the neutronic group constants are only affected by the change in density due to expansion of the fuel. The change in fuel density with temperature is implemented according to:

$$\rho(T) = \rho(T_{ref})[1 - \alpha(T - T_{ref})] \quad (1)$$

where α is thermal expansion coefficient.

The DNPs are binned into eighth groups with the decay constants and fractions shown in Table 3. The parameters are extracted using Serpent2, based on the adjoint Meulekamp's method (Zhong et al., 2011).

The fuel-filled cavity is intended to constitute a homogeneous, bare reactor and standard vacuum conditions are applied for the neutron flux at each boundary.

Throughout the benchmark the neutron flux is normalized to a fixed reference power P . An expectation is final step of the benchmark where the power transfer function is benchmarked.

Table 2
Fuel salt composition.

Isotope	Atomic fraction (%)
^6Li	2.11488
^7Li	26.0836
^9Be	14.0992
^{19}F	56.3969
$^{235}\text{U}_{235}$	1.30545

Table 3
DNP group-wise data.

Group	$\lambda_i (\text{s}^{-1})$	β_i
1	1.24667×10^{-2}	2.42173×10^{-4}
2	2.82917×10^{-2}	1.20029×10^{-3}
3	4.25244×10^{-2}	4.97887×10^{-4}
4	1.33042×10^{-1}	1.36834×10^{-3}
5	2.92467×10^{-1}	1.85916×10^{-3}
6	6.66488×10^{-1}	6.19812×10^{-4}
7	1.63478	5.79049×10^{-4}
8	3.55460	1.71090×10^{-4}

2.0.3. Input data and boundary conditions: Thermal hydraulics

The notion of a homogeneous bare reactor is reflected in the thermal hydraulics layout of the cavity as well. The thermo-physical data used in the benchmark on the CFD side is shown in Table 4.

The relative high value chosen for the Schmidt number is motivated by the need to limit mixing of DNPs while still keeping the benchmark sensitive to the individual choice of the discretization scheme. For the sake of simplicity the fuel is considered to be incompressible with a laminar flow, and changes to the density is modeled assuming the Boussinesq approximation – see Eq. (1). As such the density variations only affects the buoyancy term. In both the DTU and the SEALION approaches the thermal-hydraulics model assumes an incompressible single-phase flow with the Boussinesq approximation for the mass and momentum balance as well as a conservation of energy. Decay heat is neglected as well. In order to obtain a steady state solution in the presence of a fixed power normalization, cooling is simulated with a volumetric heat sink:

$$q'''(\mathbf{r}) = \gamma(T_{ref} - T(\mathbf{r})) \quad (2)$$

where γ denotes the volumetric heat transfer coefficient and $T(\mathbf{r})$ denotes the temperature at point $\mathbf{r}$. γ is considered to have a uniform value in the cavity but can vary from one benchmark step to another and is also allowed to vary in time. The benchmark further seeks to mimic the bare reactor by imposing a zero-velocity (i.e. no-slip) boundary condition for the fuel at all walls of the cavity except for the lid, which moves at velocity U_{lid} . The boundary condition for the DNP concentration is a homogeneous Neumann. Finally, the cavity is considered to be insulated so adiabatic boundary conditions are applied to all walls.

3. Phases and steps of the benchmark

The multiphysics benchmark is performed in three main stages: first the individual solvers for the neutronics and the thermal-hydraulics are tested, then a fully coupled steady state analysis of the model is carried out, and finally the coupling is tested for a forced convection transient scenario.

Hence, the benchmark stages are structured as follows:

Table 4
Salt thermodynamic and transport properties.

Property	Units	Value
Density	kgm^{-3}	2.0×10^3
Kinematic viscosity	m^2s^{-1}	2.5×10^{-2}
Volumetric heat capacity	$\text{Jm}^{-3}\text{K}^{-1}$	6.15×10^6
Thermal expansion coefficient	K^{-1}	2.0×10^{-4}
Prandtl number	–	3.075×10^5
Schmidt number	–	2.0×10^8

- Single physics testing: *Phase 0*
- Steady-state simulation: *Phase 1*
- Time dependent simulation: *Phase 2*

Tables 5 summarizes the input and output observables in each phase of the benchmark.

3.1. Phase 0: Single physics benchmark

Phase 0 of the benchmark is dedicated to testing of the single physics solvers. This means that the input to the given solver is fixed and the single physics solver is run until convergence is obtained. Table 5 summarizes the input and output observables in each step of phase 0.

3.1.1. Step 0.1: Velocity field

In this step the steady-state flow distribution in the cavity is evaluated assuming a fixed velocity of the lid. The aim of this step is to establish a CFD baseline flow which can be used in the next phase of the benchmark where the multi-physics coupling is introduced. The fuel is kept at the reference temperature, reducing the number of equations to be solved.

Table 5
Input and observables of phase 0–2.

Step	Input	Observables
Phase 0: Single physics benchmark		
0.1	• $U_{lid} = 0.5 \text{ ms}^{-1}$	• Velocity components along centerlines AA' and BB'.
0.2	• $T = 900\text{K}$ • $P = 1\text{GW}$	• Fission rate density, $\int_V \Sigma_f \Phi dE$ along AA' • Reactivity ρ
0.3	• Fixed flow field from Step 0.1 • Fixed heat source distribution from Step 0.2 • $\gamma = 1 \times 10^{-6} \text{ Wm}^{-3}\text{K}^{-1}$	• Temperature distribution along centerlines AA' and BB'.
Phase 1: Steady-state coupling		
1.1	• Fixed flow field from Step 0.1 • $T = 900 \text{ K}$ • $P = 1\text{GW}$	• Delayed neutron source, $\sum_i \lambda_i C_i$ along AA' and BB' • Reactivity change from Step 0.2, $\rho - \rho_{s0.2}$
1.2	• Fixed flow field from Step 0.1 • $P = 1 \text{ GW}$ • $\gamma = 1 \times 10^{-6} \text{ Wm}^{-3}\text{K}^{-1}$	• Temperature distribution along AA' and BB' • Reactivity change from Step 1.1, $\rho - \rho_{s1.1}$ • Change of fission rate density along AA' and BB' with respect to the solution obtained at Step 0.2, $\int_V \Sigma_f \Phi dE - \int_V \Sigma_{f,s0.2} \Phi_{f,s0.2} dE$
1.3	• $P = 1 \text{ GW}$ • $U_{lid} = 0 \text{ ms}^{-1}$ • $\gamma = 1 \times 10^{-6} \text{ Wm}^{-3}\text{K}^{-1}$	• Velocity components and temperature distribution along AA' and BB' • Delayed neutron source along AA' and BB' • Reactivity change from Step 0.2, $\rho - \rho_{s0.2}$
1.4	• $\gamma = 1 \times 10^{-6} \text{ Wm}^{-3}\text{K}^{-1}$ • P variable in the range [0,1] GW with a step of 0.2 GW • U_{lid} variable in the range [0,0.5] ms^{-1} , with a step of 0.1 ms^{-1}	• Reactivity change from Step 0.2, $\rho - \rho_{s0.2}$, as a function of P and U_{lid}
Phase 2: Time dependent coupling		
2.1	• Steady-state solution from Step 1.4 with $U_{lid} = 0.5 \text{ ms}^{-1}$ and $P = 1.0 \text{ GW}$ • $\gamma = 1 \times 10^{-6} \text{ Wm}^{-3}\text{K}^{-1}$	• Power gain and phase-shift as a function of the perturbation frequency.

3.1.2. Step 0.2: Neutronics

Once the flow baseline has been established by the single physics CFD solver the next step is to establish a baseline for the heating source from fission events in the cavity. As such the aim of this step is to verify and provide the fission rate density under the assumptions of a static fuel with a uniform and fixed temperature. The idea of this step is to gauge systematic effects arising from assumptions made in neutronics models as well from the choice of input data. The normalized neutron flux and the eigenvalue of the fundamental mode, namely the multiplication factor k_{eff} are verified in this step.

3.1.3. Step 0.3: Temperature

Once the steady-fluid flow and the heating source has been established, the ability of the thermal-hydraulics solver to find a steady-state solution to the temperature distribution in the cavity is assessed. Conforming with the idea of keeping the tests in the first phase of the benchmark as simple as possible this step keeps both the velocity field and the heating source fixed, thereby reducing the complexity of Navier–Stokes equation significantly and thereby allowing the benchmark to focus on the assessment of the passive scalar transport capability of the solver.

3.2. Phase 1: Steady-state coupling

Based on the steady-state solution obtained phase 0, this phase introduces the coupling gradually through a series of steps. First, the neutronics response to a fixed flow is investigated. Next, the changes to the temperature field arising from a fixed power profile is studied. The buoyancy effects will be benchmarked and finally the full coupling seek to benchmark the changes to the reactivity as a function of different power levels and cavity lid velocities.

Table 5 summarizes the input and output observables in each step of phase 1.

3.2.1. Step 1.1: Circulating fuel

The objective of this step is to assess the changes to the neutronics caused by a movement of the fuel. For the sake of simplicity and the notion of a gradually introduced coupling both the flow and the temperature distribution are kept fixed. In these settings the main effect on the neutronics should be a loss of reactivity caused by transportation of the DNPs. As such both the delayed neutron source and the reactivity change is benchmarked in this step.

3.2.2. Step 1.2: Power coupling

The objective of this step is to assess reactivity feedback mechanism caused by the changes to the fuel temperature distribution. As the temperature field changes so will the neutron flux distribution in the cavity. By fixing the velocity field and introducing a heat sink a steady-state solution can be obtained under the assumption of the fixed power normalization. Thus by assessing the steady-state temperature distribution and changes to the fission rate density the density feedback effects can be isolated and studied.

3.2.3. Step 1.3: Buoyancy

The objective of this step is to assess the capabilities of the codes to correctly model buoyancy effects caused by the temperature gradients. A natural step before benchmarking the codes in a fully coupled scenario is thus to apply the codes to a simplified setting without forced convection. Apart from the steady-state velocity field also the delayed neutron source and resulting change to the reactivity are investigated.

3.2.4. Step 1.4: Full coupling

The objective of this step is to assess the capabilities of the codes to correctly predict the loss of reactivity in a setting that constitutes a realistic reactor simulation. In this step all the effects studied in the previous steps are included, namely the changes to the neutronics caused by the fuel flow from Step 1.1, the reactivity feedback mechanism caused by the changes to the fuel temperature distribution from Step 1.2 and the buoyancy effects from Step 1.3. To mimic the simulation of a reactor under different operational condition, the simulation is repeated for different power normalization levels and cavity lid velocities.

3.3. Phase 2: Time dependent coupling

In phase 2 the full coupling developed above is studied in a transient scenario.

3.3.1. Step 2.1: Forced convection transient

The benchmark does not include a simple reactivity insertion scenario like a loss of heat sink. It is argued in (Tiberga et al., 2020) that such a comparison would be limited to the specific transient and its characteristic time constant. Instead a global perturbation in the frequency domain is applied by introducing a sine wave perturbation of the volumetric heat transfer coefficient γ . The response of a system to a sinusoidal input will in general consist of the sum of the steady-state and a transient response.

The sinusoidal perturbation to the heat transfer coefficient is implemented as a 10 % variation around the reference value of $\gamma = 1 \times 10^{-6} \text{ Wm}^{-3}\text{K}^{-1}$ with frequencies $f \in [0.0125, 0.025, 0.05, 0.1, 0.2, 0.4, 0.8] \text{ Hz}$. As a consequence of the negative fuel density feedback coefficient a sinusoidal perturbation to γ will lead to a sinusoidal shape of the reactor power, from which the power gain is defined as:

$$G(f) = \frac{(P_{\max}(f) - P_{\text{avg}}(f))/P_{\text{avg}}(f)}{(\gamma_{\max} - \gamma_{\text{avg}})/\gamma_{\text{avg}}} \quad (3)$$

and the phase-shift can be determined. Here $P_{\max, \text{avg}}(f)$ denotes the maximum and time-averaged reactor power. $\gamma_{\max, \text{avg}}$ denotes the frequency independent maximum and time-averaged value of the volumetric heat transfer coefficient. Table 5 summarizes the input and output observables in each step of phase 2.

4. Code packages

4.1. Serpent2 to OpenFOAM coupling technique

The coupling scheme is represented in Fig. 3. Two distinct software and three distinct solvers are involved: Serpent2 software is

used for neutronics part, whereas the DNP transport and the thermal-hydraulics are calculated by the OpenFOAM software. The procedure is as follows (here explained for steady-state mode):

- Serpent2 Criticality Source Simulation (Serpent CSS) is run on the same mesh as used for thermal-hydraulics, using the Serpent2 multiphysics interface capabilities. The heat source map in the domain as well as the total neutron flux, mesh-based DNP distribution, fission cross section and the delayed neutron fractions and decay constants are derived.
- The heat source map is passed on to the thermal-hydraulics solver; in this work the standard OpenFOAM *buoyantPimpleFoam* solver with *Boussinesq* approximation is used. Fuel velocity and temperature fields are obtained.
- The converged velocity field is passed on to the DNP transport solver; this is a scalar transport solver implemented in OpenFOAM, which, using the neutronics parameters obtained from Serpent2 and the velocity field obtained from OpenFOAM, solves the scalar transport equation for each DNP group:

$$\frac{\partial C_i}{\partial t} = -\nabla(uC_i) + \nabla\left(\frac{v_t}{Sc'}\right)\nabla C_i - \lambda_i C_i + \beta_i \Phi v \Sigma_f \quad (4)$$

where C_i is the DNP concentration per group, u is the flow velocity, v_t is the turbulent kinematic viscosity, Sc' is the turbulent Schmidt number, λ_i is the decay constant per group, β_i is the fractional yield of delayed neutrons to fission neutrons per group, Φ is the neutron flux and Σ_f is the fission cross section. The converged DNP distributions are written out for each group.

- The temperature distribution from the thermal-hydraulics solver and the DNP sources from the DNP transport solver are fed back into Serpent and the criticality source simulation routine is repeated.

This routine is repeated as many times as necessary to ensure that the results have all converged. For the transient scheme, the exchange of coupling terms is performed within each time-step. An advantage of this multiphysics scheme is that it is simple to implement and it also simplifies the debugging process. Another advantage is the flexibility of the concept, e.g. one can choose to use either mesh-wise DNP distributions or, for more complicated geometries, the total group-wise concentration can be used, with proper spatial weighting of DNPs, which will eliminate the necessity to use a mesh with the same number of cells in the neutronics module. Moreover, the fuel velocity field impacts the neutronics module as an external operator, via the updated DNPs source which means that no modifications as such are required in the neutronics solvers. The main disadvantages of the method are the computational time needed for the Monte-Carlo calculations and the need to use small time-steps in the time dependent simulations to limit the errors assigned from the coupling terms.

The approach in the transient phase of the benchmark is essentially the same as described above. This approach is justifiable provided that the thermal-hydraulic time-steps are small enough, during a time step.

Note that utilizing Serpent2 in CSS mode rather than transient mode implicitly is equivalent to solving the transport problem in the zero prompt lifetime approximation where the reactor period is solely determined from the delayed neutrons. Such an approximation is only valid for transients with characteristic time scales much larger than the prompt neutron lifetime and therefore only valid for moderate reactivity insertions and low fuel velocities. Since the main focus of the Serpent-OpenFOAM in the present work is to replicate the steady-state of the benchmark, this approximation is justified.

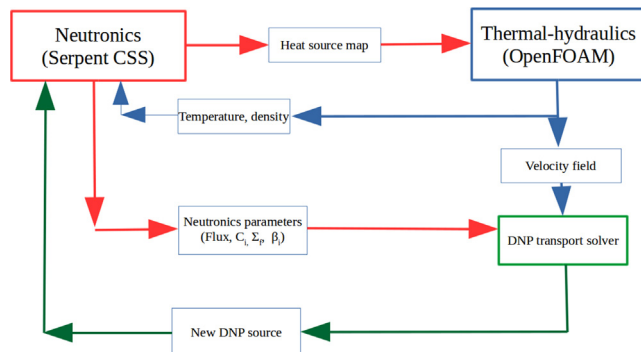


Fig. 3. The multiphysics coupling scheme.

4.1.1. Specifications for benchmark calculations

A mesh dependence resulted in the 80×80 mesh being selected as a baseline for the simulations. As a criteria for checking the mesh independence, the average velocity in the domain is chosen and the results are reported in Fig. 4.

Using the same mesh for both thermal-hydraulics and neutronics simulations allows for direct exchange of variables between the two modules, e.g. the volumetric heat source and the DNP distributions can be generated and used without any modifications related to mesh translations. The OpenFOAM generated mesh is linked to Serpent2, used for the neutronics simulations, by a dedicated multiphysics interface (Tuominen et al., 2016).

The discretization schemes for the integration of the temporal, diffusive and convective terms are a CrankNicolson, linear Gaussian and a linear limited Gaussian, respectively. The convergence criteria for the coupled solver is influenced by the convergence criteria for the mono-physics solvers individually. For the neutronics solver the number of active and inactive cycles are chosen to be high enough to ensure convergence in the Serpent CSS run. On the thermal-hydraulics a tolerance of 10^{-8} to 10^{-4} is used in OpenFOAM depending on the quantity is being solved for. For the DNPs the tolerance was set to 10^{-8} whereas for all other quantities, the tolerance was set to 10^{-4} . For single quantities a relative tolerance was set. In steady-state simulations, the codes are iterated until convergence.

4.2. SEALION framework description

In general, point kinetics solvers are expected to capture the physics of a reactor system to a good approximation for a given reactor transient, as long as the transient is mainly driven by the fundamental form. If the fundamental form, and thus the fission power shape, does not change appreciably during the transient, the results obtained from a point kinetics model can be directly convoluted with the shape function of the fission power, thus allowing for extracting the time dependent local power density. In the case of coupled simulations, this procedure can be carried out iteratively in a suitable coupling scheme. In this sense, for coupled analysis, the point kinetics approximation is expected to be less suited for transients where the fundamental form changes significantly (e.g. simulation of rod ejection) and where higher order methods, e.g., diffusion models, are required. Nevertheless, due to

the extensive application of point kinetics solvers in transient reactor analysis, as well as the extensive regulatory experience and confidence with such solvers, it is attractive to examine to what an extend the reduced order physics point kinetics framework can be applied to liquid fueled reactor systems (Valocchi et al., 2020). The adaptation of point kinetics methods to describe a molten salt reactor is not a new idea. There has been extensive effort put into the subject starting from a first model proposed by ORNL almost 60 years ago. These efforts include the development of purely point-kinetics models (in various forms); quasi-static models; mixed models with point-kinetics power and spatial DNPs; various approaches (lumped, weighted-lumped) for weighting the reactivity feedbacks; various approaches for weighting the DNPs (Kerlin and Ball, 1965; Lapenta et al., 2001; Guerrieri et al., 2013; Zanetti et al., 2015; Dulla et al., 2004; Guerrieri et al., 2004; Cammi et al., 2004; Krepel et al., 2005; Fiorina et al., 2010). In that sense, the SEALION framework is a new implementation of well know concepts and should be seen as a contribution to the list of studies carried out in the field.

The main challenge in modifying point kinetics to accommodate liquid fueled reactor systems is the change in neutron importance DNPs. In general, in a static fueled system, i.e. for most conventional reactors, a neutron resulting from decay of a DNP created at some location will have the same importance as a prompt neutron appearing in the same location, disregarding neutron emission spectra effects. In effect, the equations governing the dynamics can be collapsed into the point kinetics equations for the reactor and DNPs amplitudes appearing as a set of ODEs in time with no reference to spatial variables (see e.g. Diniz et al., 2020) with DNPs only distinguished by the fact that they appear in a "delayed" manner and with emission spectrum importance effects captured by the effective delayed neutron fractions. The point kinetics equations can therefore be seen as an effective theory in the sense that information about the spatial distribution of the fundamental form/shape function has been lost, but appears indirectly in terms of the effective kinetics parameters appearing in the resulting equations.

In the case of liquid fueled reactors such an approach is bound to fail. Indeed, suppose that a DNP is created near the center of the reactor core and subsequently moves to the edge of the core before decaying. Such a DNP will clearly appear with decreased importance than had it appeared statically. This effect needs to be accounted for in a satisfactory point kinetics model and clearly spatial information about the DNP concentrations will be needed to compute the importance. In this sense, we have no real hope of constructing a true point kinetics model for a liquid fueled system as spatial information about the DNPs will be needed to predict dynamics. However, given that the delayed neutron fraction is small the bulk part of neutron multiplicity is driven by prompt neutrons; the introduction of moving DNPs is not expected to change the fundamental and adjoint form of the system significantly. This expectation is also supported by theoretical arguments presented in (Ash, 1979). It is therefore reasonable to expect that a modified point kinetics approximation can still be employed in the sense that the equations retain the usual form, but the DNP contribution acquires the equation for the reactor amplitude from a non-trivial spatial weighting over the DNP concentrations - see below for details. In particular, the reactivity appearing in the point kinetics equations is related to the k -eigenvalue and can, for a given temperature distribution, be extracted using Monte-Carlo methods. As it is undesirable to perform a time consuming full k -code calculation in each time step of a coupled calculation, the SEALION framework employs a linear separable reactivity model on a pre-defined neutronics mesh for evaluating the reactivity during the coupled simulation. The input parameters of this model are indi-

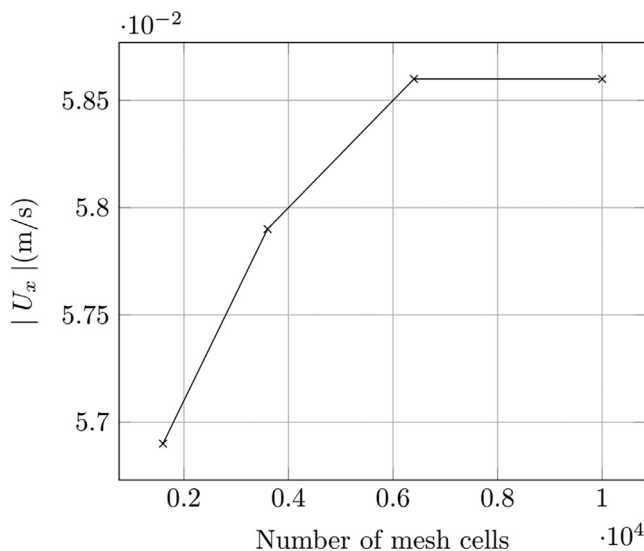


Fig. 4. Results from a mesh dependence study.

vidual reactivity responses for the neutronics mesh cells under temperature perturbations which are evaluated from a Monte-Carlo k -code simulation, here using Serpent2. Using this model, we can derive the reactivity response of a given hydrodynamic configuration from temperature data on the neutronics mesh. In particular, the model takes local temperature effects into account expected to be of high importance for liquid fueled systems. Liquid fueled reactor are expected to exhibit larger local temperature effects in the fuel when compared to their solid fueled counterparts. This is due to the, in general, more pronounced thermal expansion and associated temperature coefficients as well as the direct coupling to the flow of the liquid fuel.

Finally, since the fundamental form is not expected to change significantly, the shape function and the fission power deposition shape are taken to be fixed parameters in the current model with both quantities extracted from a single high precision k -code calculation. The shape function is used to weight the DNP contribution while the fission power shape is used to compute the local deposited heat during each time step of the coupled simulation.

The PK module is the first of three central SEALION modules. The three modules cover:

- **PK module:** This neutronics module is based on the modified point-kinetics approach and is capable of describing the evolution of neutronics parameters, calculating temperature and density feedback coefficients and power evolution. It is also capable of calculating mesh-based delayed neutron precursor transport and concentrations.
- **nCFD module:** An in-house developed solver based on the open source platform OpenFOAM with built-in capabilities of DNP, coolant, fuel and heat transport models.
- **Master iterator (MI) module:** Integrates and couples the PK and nCFD modules. The MI modules handles dynamic time stepping, geometric mesh mapping, operational state specification (steady-state and transient).

In the coupled tool, the PK module will receive temperature and DNP distributions from the nCFD module. Using a reactivity map produced prior to the coupled calculations with a Monte Carlo neutronics code, it will then calculate, among other things, the distribution and evolution of power and neutron flux, which will be read and analysed to produce an updated set of input parameters for the nCFD module. This data analysis and exchange will be controlled by the MI module. The full simulation process flowchart is shown in Fig. 5.

The software modules mentioned above are all envisioned to undergo a qualification. However at the current stage only nCFD module is foreseen to undergo that process in the near future.

The starting point for the SEALION computational framework are the general time-dependent point kinetics equations. Expressed in terms of reactor power P and DNP concentration C_I in a given group I the point kinetics equations looks like:

$$\dot{P} = \left(\frac{\rho - \beta}{\Lambda_{(p)}} \right) P + \sum_{I=1}^{N_{DNP}} \lambda_I C_I \quad (5)$$

$$\dot{C}_I = \left(\frac{\beta_I}{\Lambda_{(p)}} \right) P - \lambda_I C_I \quad (6)$$

where N_{DNP} denotes the number of DNP groups which in the current study is set to eight. $\Lambda_{(p)}$ denotes the prompt neutron generation time, β_I the relative fraction of delayed neutrons from DNP group I , $\beta = \sum_{I=1}^8 \beta_I$ is the total delayed neutron fraction and λ_I the decay constant of DNP group I . Finally, ρ (the reactivity) represents the branching of the fission chains and the amount of multiplication in the system.

Simulation prep.

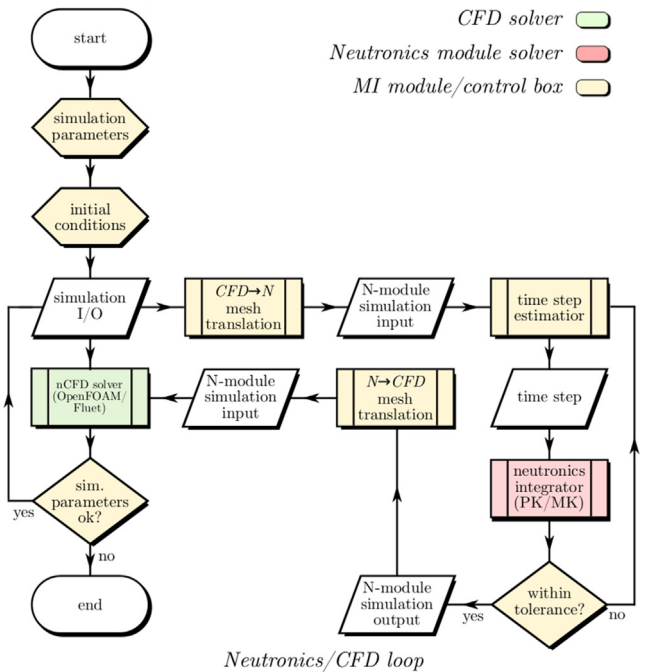


Fig. 5. Full simulation flowchart.

In order to incorporate neutron importance effects, the DNP coupling term is convoluted with the adjoint flux, which in turn is approximated by the flux, consistent with a simple one-group approximation, and finally recasted on a mesh:

$$|\dot{\mathbb{P}}\rangle = \left(\frac{\rho - \beta}{\Lambda_{(p)}} \right) |\mathbb{P}\rangle + \sum_{I=1}^{N_{DNP}} (\lambda_I \mathcal{P} \otimes \text{Vol} \psi) |\mathbb{C}_I\rangle \quad (7)$$

$$|\dot{\mathbb{C}}_I\rangle = \left(\frac{\beta_I}{\Lambda_{(p)}} \right) |\mathbb{P}\rangle - \lambda_I |\mathbb{C}_I\rangle \quad (8)$$

Here $|\mathbb{P}\rangle$ and $|\mathbb{C}_I\rangle$ denote the mesh state vectors for the power density and the DNP concentration. $\text{Vol} \psi$ denotes an inner product between two vectors Vol and ψ that contains the volumes and neutron flux of each individual mesh cell. $\mathcal{P}$ denotes the power shape unit normalized in a L_2 sense.

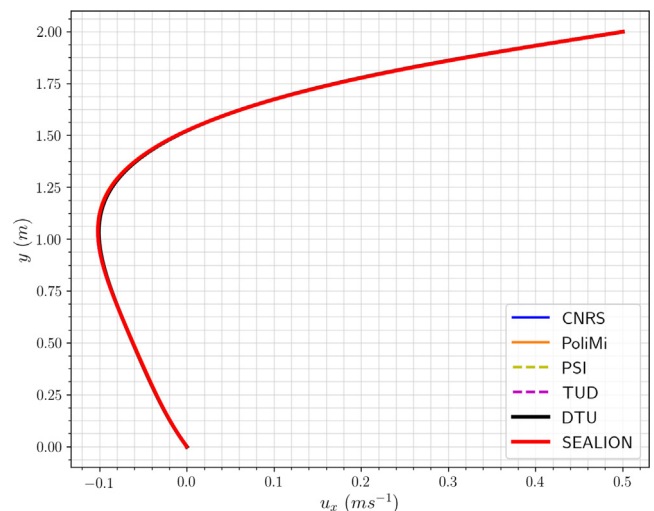


Fig. 6. Step 0.1 – Horizontal velocity component along BB' .

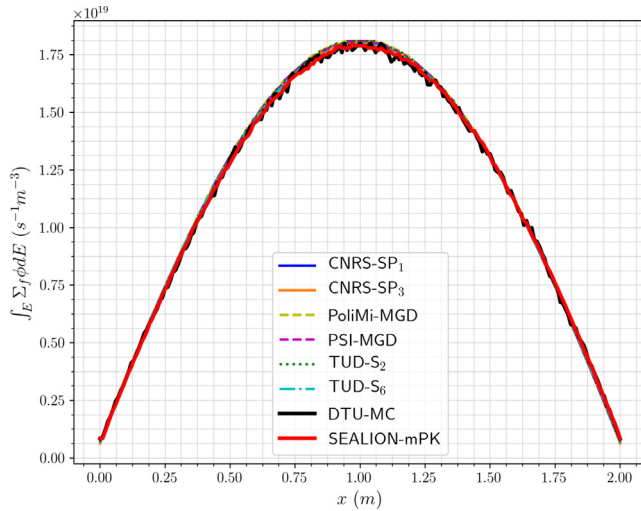


Fig. 7. Step 0.2 – Fission rate density along AA'.

Table 6
Step 0.2 – System excess reactivity.

Code	ρ (pcm)
DTU	740 ± 6
SEALION	741.2 ± 2.5
CNRS-SP ₁	411.3
CNRS-SP ₃	353.7
PoliMi	421.2
PSI	411.7
TUD-S ₂	482.6
TUD-S ₆	578.1

The deterministic approaches for computing the point-kinetics parameters with the explicit solution for the adjoint flux is considered a difficult task in continuous energy Monte Carlo codes. For these reasons, other approaches (that do not require explicit computation of the adjoint flux) are often employed. In the SEALION framework, the iterated fission probability (IFP) approach within Serpent2 is used to extract values for the point-kinetics parameters (Leppänen et al., 2014).

The SEALION code relies on data exchange between the CFD and neutronics modules, which in general means employing different meshes. The neutronics mesh is expected to be somewhat coarser than the mesh employed by the CFD software for computational reasons. In order to handle data exchange between the SEALION modules, data on one mesh therefore needs to be translated to the other mesh and vice versa. In general, mesh translation from a fine to a coarse mesh is performed by a weighted average algorithm while mesh translations from a coarse to a fine mesh will require an interpolation algorithm that assumes that the characteristic length scale for changes in the relevant physical quantities are small (compared to the meshing length).

The SEALION framework implements an algorithm based on the `scipy.interpolate.griddata` (Jones et al., 2001) algorithm for all coarse-to-fine mesh translations. Other methods like the weighted average projection method can be used as well. A dedicated module has been developed to perform mesh translations in a systematic way in order to ensure consistency between various meshes used by the SEALION code. The module is based on the OpenFOAM polyMesh format as well as the python packages NumPy (Oliphant, 2006) and SciPy (Jones et al., 2001). The first python package is used for the manipulation of multi-dimensional arrays and the latter for interpolation and weighted average algorithm.

4.2.1. Specifications for benchmark calculations

The mesh for computing the reactivity response is a coarse structured uniform mesh containing 10×10 cells, while the mesh used for the neutronics solver is a finer mesh containing 20×20 cells, likewise structured and uniform. The mesh used for the CFD solver is a structured non-uniform mesh containing 200×200 cells. Mesh convergence was ensured, similarly to the DTU coupling approach (Fig. 4).

The input data needed to the modified point kinetics equations are extracted using Serpent2.1.31 (Leppänen et al., 2014). The power density is likewise extracted using Serpent2 and translated into mesh state vectors for the power density $|P\rangle$ using the dedicated mesh translation module.

On the thermo-hydraulics side the SEALION code employs an Euler, corrected linear Gaussian and an upwind Gaussian discretization scheme for the integration of the temporal, diffusive and convective terms, respectively.

The convergence criteria for the coupled solver is influenced by the convergence criteria for the mono-physics solvers individually. The neutronics solver uses a SciPy implementation of explicit Runge-Kutta method of order 5(4) (Dormand and Prince, 1980). The convergence is therefore controlled assuming accuracy of the fourth-order method, but steps are taken using the fifth-order accurate formula. On the thermal-hydraulics a tolerance of 10^{-8} to 10^{-6} was used in OpenFOAM, depending on the quantity is being solved for. For the hydro-static pressure the tolerance was set to 10^{-8} whereas for all other quantities, the tolerance was set to 10^{-6} . For single quantities a relative tolerance can also be set. In steady-state simulations, the codes are iterated until convergence.

Transient calculations were performed taking a time step equal to 1/200 of the perturbation period.

5. Benchmark results

In this section the main results are presented and compared to the benchmark analysis (Tiberga et al., 2020). For the sake of transparency the step numbers in the figures and tables refer to the step numbering used in the benchmark analysis.

5.0.2. Assessing the results

Given the nature of the benchmark no reference solution exists, however, as a measure of any discrepancy of any individual code to the average result, we employ a χ^2 metric,

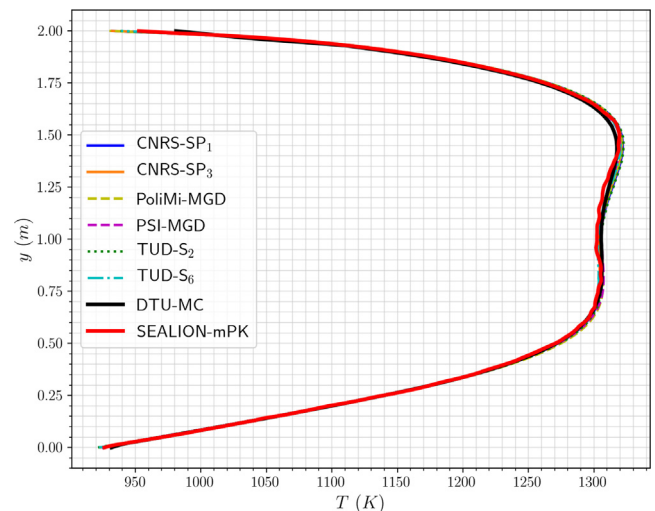


Fig. 8. Step 0.3 – Temperature along BB'.

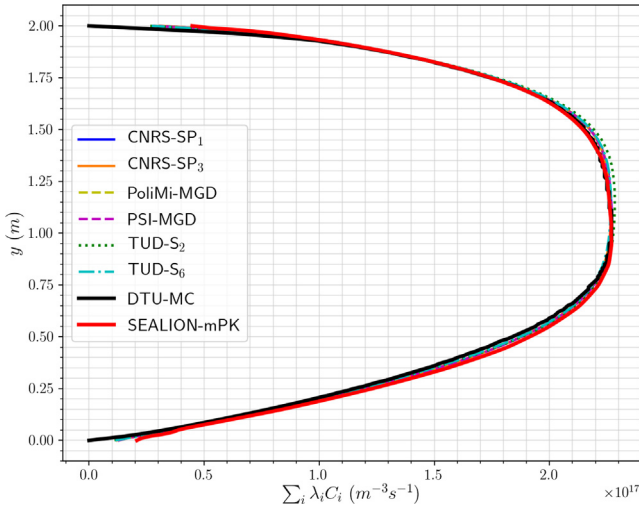


Fig. 9. Step 1.1 – Delayed neutron source along BB' .

Table 7
Step 1.1 – Reactivity change from Step 0.2.

Code	$\rho - \rho_{s0.2}$ (pcm)
DTU	-60 ± 13
SEALION	-57.6 ± 8.6
CNRS-SP ₁	-62.5
CNRS-SP ₃	-62.5
PoliMi	-62.0
PSI	-63.0
TUD-S ₂	-62.0
TUD-S ₆	-62.7

$$\epsilon_C = \sqrt{\frac{\sum_{i=1}^{N_p} (Q_C(\mathbf{r}_i) - Q_{avg}(\mathbf{r}_i))^2}{\sum_{i=1}^{N_p} Q_{avg}^2(\mathbf{r}_i)}} \quad (9)$$

where $\mathbf{r}_i$ denotes a point on the sample lines, $N_p = 201$ is the number of points used in sampling of the quantity Q_C for a given code C . The average value $Q_{avg}(\mathbf{r}_i) = \frac{1}{N_C} \sum_{c=1}^{N_C} Q_c(\mathbf{r}_i)$, however excludes the code under scrutiny. A collective measure for the full set of codes is taken to be the averaged value of ϵ_C : $\epsilon = \frac{1}{N_C} \sum_{C=1}^{N_C} \epsilon_C$.

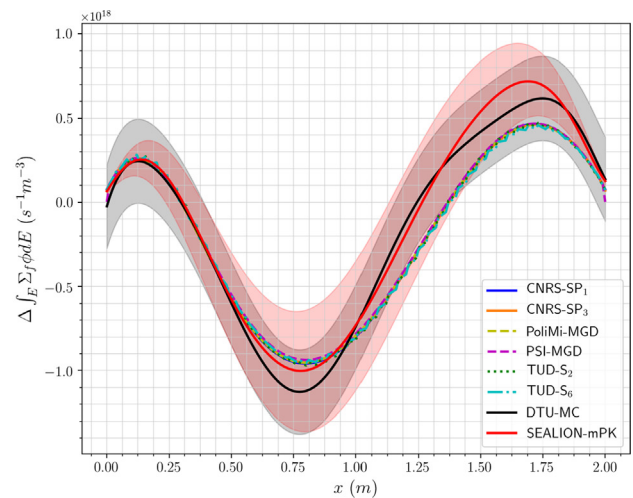
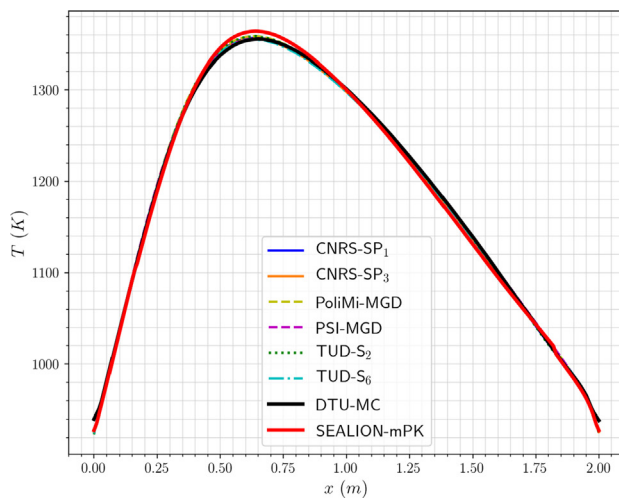


Fig. 10. Step 1.2 – Temperature and change of fission rate density along AA' . The colored band around the curves show the statistical uncertainty.

5.1. Phase 0: Single physics benchmark

5.1.1. Step 0.1: Velocity field

The first test of the single physics solvers is the steady-state velocity field of the fuel given a fixed lid velocity: $U_{lid} = 0.5 \text{ ms}^{-1}$.

In Fig. 6, the horizontal velocity component along the cavity vertical centerline is shown, providing a good agreement between the results of this work and the benchmark participants.

The code-wise discrepancy ϵ_C (DTU) varies between 0.4% and 1.3% along AA' centerline and between 0.3% and 0.8% along BB' centerline. For the SEALION code the corresponding numbers are ϵ_C (SEALION) $\in [0.4\%, 0.8\%]$ and ϵ_C (SEALION) $\in [0.3\%, 0.7\%]$. Respectively, these numbers are to be compared to an average relative variation of 0.35% for velocity profiles along AA' and 0.8% for the BB' profiles for the benchmark participants. A statistical test based on the measure ϵ_C yields that both the DTU and SEALION code is statistically compatible with the rest of the benchmark participants (p-value ≈ 1). A Kolmogorov–Smirnov test yields the same result. The zero velocity of both codes on the boundaries reflects the no-slip condition imposed on all boundaries except the lid which moves at and 0.5 ms^{-1} (Ghia et al., 1982; Botella and Peyret, 1998).

The velocity values used for the plots are provided in Table A.11 in Appendix A and are compared for different benchmark steps in Fig. B.13 in Appendix B.

5.1.2. Step 0.2: Neutronics

The objective of Step 0.2 is to establish a baseline for the heating source from fission events in the cavity and thereby provide a test of the neutronics module. Fig. 7 shows the fission rate density and Table 6 shows the excess reactivity.

A good agreement with the benchmark is observed both for the SEALION and for the DTU codes. The code-wise discrepancy is ϵ_C (DTU) = 1.6% and ϵ_C (SEALION) = 1.3% which is at the level of the statistical uncertainty in the individual sample bins. These numbers are to be compared to a code-wise discrepancy for benchmark participants in the range of 0.2%–0.6% and an average relative deviation below 0.4%. At this stage both codes relies on Serpent2 solely to extract the fission rate density. As such it is expected that the two codes give similar results. To ensure that the fission rate density is tallied in a region similar to the deterministic codes a relative small detector volume is used. As a consequence hereof relatively large fluctuations can be observed in the results for both the DTU and SEALION code. It should be emphasized that the statistical

Table 8
Step 1.2 – Reactivity change from Step 1.1.

Code	$\rho - \rho_{s1.1}$ (pcm)
DTU	-1094.0 ± 103
SEALION	-1217.5 ± 30
CNRS-SP ₁	-1152.0
CNRS-SP ₃	-1152.7
PoliMi	-1161.0
PSI	-1154.8
TUD-S ₂	-1145.2
TUD-S ₆	-1122.0

fluctuations present in Fig. 7 will not propagate into the rest of the benchmark as the input to the neutronics solvers relies on the fission rate density extracted on the full volume and not the subset used for tallying. Also note here that both the DTU and SEALION results gives significantly higher values at the boundaries ($x = 0$ m and $x = 2$ m) when compared to the average of the results from the other codes. This is expected due to the difference in the implementation of the vacuum boundary conditions between diffusion and higher order transport discretizations. As can be seen from Table A.12 in Appendix A, DTU and SEALION results show a better agreement with the higher order codes as expected due to the multi-group and higher order nature of Monte-Carlo codes. A statistical test similar to the one performed in Step 0.1 implies that the hypothesis that the fission rate density distribution from the

DTU and SEALION codes stems from the same distribution as the benchmark participants can not be rejected.

Table 6 gives the system excess reactivity for the various codes. The DTU and SEALION codes both yield significantly larger excess reactivity than the other codes. The explanation for the discrepancy is twofold. First, the more advanced transport models are expected to give different results from the diffusion codes. In comparison the DTU and SEALION codes differ from the average of the other participants by 300 pcm. Another factor contributing to the discrepancy is the different choice of nuclear libraries. The difference in excess reactivity when evaluated with Serpent2 using JEFF3.1 vs JEFF3.2 is 386 pcm which is consistent with the difference observed between the benchmark participants and the DTU and SEALION codes. Further studies have been carried out to estimate the systematic uncertainties arising from the specific choice of nuclear library. Differences in the fission rate density from the different choice of nuclear library are locally below 0.2% and thereby compatible with the code-wise and average relative discrepancy for benchmark participants. The study of systematic uncertainties have also be extended to include some of the later steps in the benchmark. Here the uncertainties from a particular choice of nuclear library fall below the code-wise and average relative discrepancy reported by the benchmark participants. This picture is consistent with the notion that discrepancies between the benchmark participants reported in Table 6 does not propagate to the rest of the benchmark.

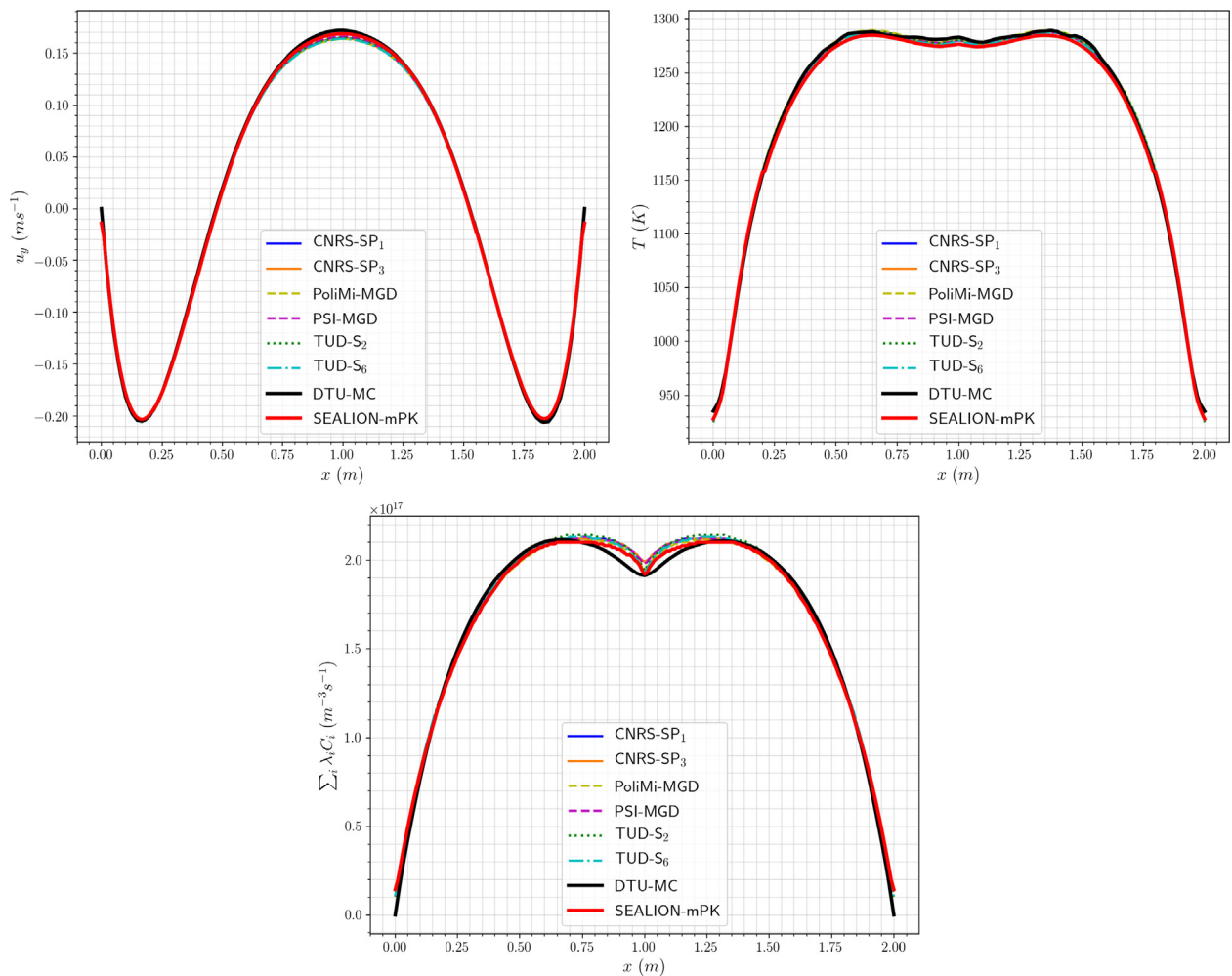


Fig. 11. Step 1.3 – Vertical velocity component, temperature, and delayed neutron source along AA'.

Table 9
Step 1.3 – Reactivity change from Step 0.2.

Code	$\rho - \rho_{s0.2}$ (pcm)
DTU	-1249 ± 25
SEALION	-1218.2 ± 22
CNRS-SP ₁	-1220.5
CNRS-SP ₃	-1220.5
PoliMi	-1227.0
PSI	-1219.6
TUD-S ₂	-1208.5
TUD-S ₆	-1184.4

5.1.3. Step 0.3: Temperature

In this step the fixed velocity field from Step 0.1 and the fixed volumetric heat map from Step 0.2 are used to test the ability of the codes to predict the temperature of the fuel. The temperature profile along BB' is shown in Fig. 8. The code-wise discrepancy is $\epsilon_c(DTU) \in [0.2\%, 0.4\%]$ and $\epsilon_c(SEALION) \in [0.2\%, 0.6\%]$ along AA' and BB' centerlines. These numbers are to be compared to an average relative discrepancy below 0.1%. A good agreement between this work and the benchmark participants results is observed. It should be noted that the results from the DTU and SEALION codes presents slight deviations from the average of the results from the other participants. However, the deviations are statistically insignificant.

Table 10
Step 1.4 – Reactivity change from Step 0.2 as a function of P and U_{lid} .

Code	$U_{lid}(ms^{-1})$	$\rho - \rho_{s0.2}$ (pcm)				
		P (GW)				
		0.2	0.4	0.6	0.8	1.0
DTU	0.0	-206 ± 28	-440 ± 30	-728 ± 42	-913 ± 27	-1118 ± 29
SEALION		-269.6 ± 22	-523.3 ± 29	-770.1 ± 27	-1013.2 ± 24	-1253.71 ± 26
CNRS-SP ₁		-264.5	-503.4	-737.8	-974.2	-1220.5
CNRS-SP ₃		-265.8	-503.4	-738.5	-976.2	-1220.7
PoliMi		-266.0	-498	-734	-975	-1227
PSI		-268.0	-504.8	-739	-975	-1215.1
TUD-S ₂		-263.7	-498.1	-731.1	-967.2	-1208.5
TUD-S ₆		-258.0	-487.8	-716.3	-947.9	-1184.4
DTU	0.3	-265 ± 28	-451 ± 32	-671 ± 30	-850 ± 29	-1098 ± 29
SEALION		-260.7 ± 26	-512.1 ± 24	-757.7 ± 27	-1000 ± 24	-1239.8 ± 26
CNRS-SP ₁		-269.5	-502.9	-735.2	-970.9	-1212.1
CNRS-SP ₃		-269.8	-503.5	-735.3	-971.0	-1212.4
PoliMi		-278.0	-503.0	-734.0	-970.0	-1219.0
PSI		-274.0	-504.8	-735.6	-969.1	-1206.8
TUD-S ₂		-269.5	-498.1	-727.7	-961.2	-1200.2
TUD-S ₆		-263.8	-488.0	-713.2	-942.2	-1176.4
DTU	0.5	-289 ± 30	-468 ± 27	-655 ± 28	-836 ± 31	-1093 ± 33
SEALION		-254.4 ± 28	-503.2 ± 24	-746.1 ± 30	-985.9 ± 26	-1223.1 ± 27
CNRS-SP ₁		-276.5	-503.5	-732.9	-966.3	-1204.8
CNRS-SP ₃		-276.8	-503.5	-733	-966.5	-1205.2
PoliMi		-284	-508	-737	-972	-1214
PSI		-278.1	-504.6	-733.1	-964.3	-1199.8
TUD-S ₂		-273.1	-497.8	-725.2	-956.4	-1193
TUD-S ₆		-267.5	-487.8	-710.8	-937.6	-1169.7

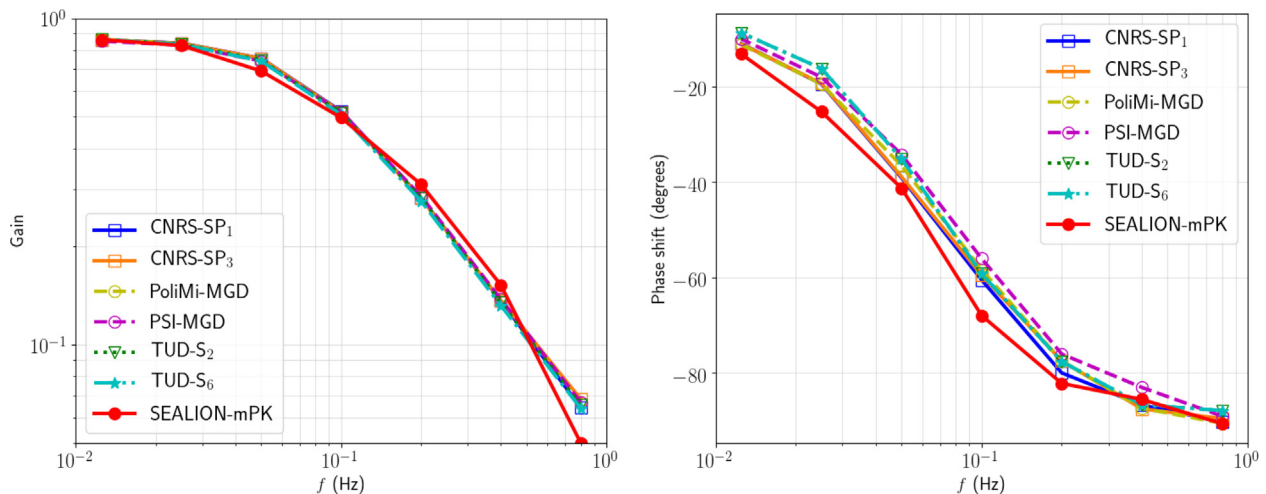


Fig. 12. Step 2.1 – Bode diagrams of power gain and phase-shift as a function of the frequency of the γ wave.

The values for temperature along the two centerlines are given in Table A.13 in Appendix A. Fig. B.14 in Appendix B shows the temperature distributions for various benchmark steps.

5.2. Phase 1: Steady-state coupling

5.2.1. Step 1.1: Circulating fuel

As mentioned in Section 3.2.1 the goal of this step is to investigate the reactivity loss due to the transport of delayed neutron precursors. In order to isolate the effect a simplified version of the coupling is introduced where both the flow field and the temperature distribution are kept fixed. Fig. 9 shows the total delayed neutron source along the vertical centerline. The agreement with the benchmark results is satisfactory, with a code-wise discrepancy of $\epsilon_c(DTU) \in [2.4\%, 2.8\%]$ and $\epsilon_c(SEALION) \in [1.2\%, 3.1\%]$ along the AA' and BB' centerlines. These numbers are to be compared to a code-wise discrepancy for benchmark participants of 0.2%–0.7% and an average relative discrepancy below 0.3%. The raw values for the delayed neutron precursor source are provided in

Table A.14 in Appendix A. Figs. B.15 and B.16 in Appendix B show the delayed neutron precursor distributions for various benchmark steps for the first and fifth family.

Table 7 summarizes the reactivity loss with respect to the static case presented in Step 0.2. The results from the DTU and SEALION codes both differ from the average of the other participants by 2–7 pcm. The objective of this step in the benchmark is to assess the capabilities of the codes to correctly model the reactivity loss due to the transport of delayed neutron precursors. A discrepancy in the either the fission rate density or the steady state flow field will propagate into the results for the reactivity loss/gain. However, the results are compatible with the benchmark participants within the uncertainties.

5.2.2. Step 1.2: Power coupling

The impact of a non-uniform temperature on the flux and in turn the effect of the flux deformation on the temperature is evaluated. The temperature impact on the reactivity is assessed via the fuel density feedback.

Table A.11
Step 0.1 – Velocity components along centerlines AA' and BB'.

Observable	Code	Results along AA' (points coordinates are expressed in m)								
		(0,0,1)	(0.25,1)	(0.5,1)	(0.75,1)	(1.0,1)	(1.25,1)	(1.5,1)	(1.75,1)	(2.0,1)
u_x (ms ⁻¹)	DTU	0.000E+00	-1.939E-02	-5.358E-02	-8.281E-02	-1.008E-01	-1.023E-01	-7.853E-02	-3.084E-02	0.000E+00
	SEALION	0.000E+00	-1.929E-02	-5.367E-02	-8.335E-02	-1.018E-01	-1.034E-01	-7.913E-02	-3.072E-02	0.000E+00
	CNRS	0.000E+00	-1.924E-02	-5.372E-02	-8.369E-02	-1.025E-01	-1.043E-01	-7.972E-02	-3.080E-02	0.000E+00
	PoliMi	0.000E+00	-1.922E-02	-5.365E-02	-8.357E-02	-1.023E-01	-1.041E-01	-7.947E-02	-3.066E-02	0.000E+00
	PSI	0.000E+00	-1.929E-02	-5.366E-02	-8.332E-02	-1.018E-01	-1.034E-01	-7.912E-02	-3.072E-02	0.000E+00
	TUD	1.002E-06	-1.922E-02	-5.372E-02	-8.371E-02	-1.025E-01	-1.044E-01	-7.977E-02	-3.081E-02	4.198E-06
u_y (ms ⁻¹)	DTU	0.000E+00	7.279E-02	8.506E-02	5.975E-02	1.211E-02	-4.683E-02	-9.465E-02	-8.753E-02	0.000E+00
	SEALION	0.000E+00	7.254E-02	8.526E-02	6.021E-02	1.232E-02	-4.732E-02	-9.529E-02	-8.711E-02	0.000E+00
	CNRS	0.000E+00	7.266E-02	8.575E-02	6.084E-02	1.251E-02	-4.789E-02	-9.606E-02	-8.722E-02	0.000E+00
	PoliMi	0.000E+00	7.139E-02	8.433E-02	6.007E-02	1.269E-02	-4.691E-02	-9.472E-02	-8.621E-02	0.000E+00
	PSI	0.000E+00	7.265E-02	8.534E-02	6.021E-02	1.230E-02	-4.734E-02	-9.536E-02	-8.720E-02	0.000E+00
	TUD	5.877E-06	7.269E-02	8.580E-02	6.089E-02	1.252E-02	-4.794E-02	-9.613E-02	-8.726E-02	-1.013E-05
Observable	Code	Results along BB' (points coordinates are expressed in m)								
		(1,0,0)	(1,0.25)	(1,0.5)	(1,0.75)	(1,1,0)	(1,1.25)	(1,1.5)	(1,1.75)	(1,2,0)
u_x (ms ⁻¹)	DTU	0.000E+00	-3.530E-02	-6.223E-02	-8.629E-02	-1.008E-01	-8.611E-02	-1.123E-02	1.696E-01	5.000E-01
	SEALION	0.000E+00	-3.504E-02	-6.209E-02	-8.664E-02	-1.018E-01	-8.739E-02	-1.196E-02	1.705E-01	5.000E-01
	CNRS	0.000E+00	-3.517E-02	-6.242E-02	-8.720E-02	-1.025E-01	-8.766E-02	-1.147E-02	1.717E-01	5.000E-01
	PoliMi	0.000E+00	-3.423E-02	-6.107E-02	-8.613E-02	-1.023E-01	-8.861E-02	-1.299E-02	1.706E-01	5.000E-01
	PSI	0.000E+00	-3.511E-02	-6.217E-02	-8.667E-02	-1.018E-01	-8.731E-02	-1.191E-02	1.705E-01	5.000E-01
	TUD	1.494E-06	-3.519E-02	-6.244E-02	-8.724E-02	-1.025E-01	-8.770E-02	-1.146E-02	1.718E-01	5.000E-01
u_y (ms ⁻¹)	DTU	0.000E+00	1.077E-04	7.568E-04	3.921E-03	1.211E-02	2.411E-02	2.928E-02	1.478E-02	0.000E+00
	SEALION	0.000E+00	7.850E-05	6.866E-04	3.882E-03	1.232E-02	2.475E-02	2.996E-02	1.482E-02	0.000E+00
	CNRS	0.000E+00	5.641E-05	6.309E-04	3.862E-03	1.251E-02	2.524E-02	3.048E-02	1.500E-02	0.000E+00
	PoliMi	0.000E+00	9.118E-05	7.484E-04	4.046E-03	1.269E-02	2.534E-02	3.050E-02	1.500E-02	0.000E+00
	PSI	0.000E+00	7.727E-05	6.822E-04	3.875E-03	1.230E-02	2.472E-02	2.994E-02	1.481E-02	0.000E+00
	TUD	1.501E-06	5.260E-05	6.209E-04	3.853E-03	1.252E-02	2.528E-02	3.053E-02	1.502E-02	7.987E-06

Table A.12
Step 0.2 – Fission rate density along AA'.

Observable	Code	Results along AA' (points coordinates are expressed in m)								
		(0,0,1)	(0.25,1)	(0.5,1)	(0.75,1)	(1,0,1)	(1.25,1)	(1.5,1)	(1.75,1)	(2,0,1)
$\int_E \Sigma_f \phi dE$ (s ⁻¹ m ⁻³)	DTU-MC	8.800E+17	7.410E+18	1.280E+19	1.660E+19	1.790E+19	1.660E+19	1.300E+19	7.560E+18	8.800E+17
	SEALION-mPK	8.510E+17	7.320E+18	1.280E+19	1.660E+19	1.790E+19	1.660E+19	1.310E+19	7.590E+18	8.510E+17
	CNRS-SP ₁	6.900E+17	7.440E+18	1.310E+19	1.680E+19	1.810E+19	1.680E+19	1.310E+19	7.440E+18	6.900E+17
	CNRS-SP ₃	6.210E+17	7.450E+18	1.300E+19	1.670E+19	1.800E+19	1.670E+19	1.300E+19	7.450E+18	6.210E+17
	PoliMi-MGD	7.780E+17	7.470E+18	1.310E+19	1.680E+19	1.810E+19	1.680E+19	1.310E+19	7.470E+18	7.780E+17
	PSI-MGD	8.620E+17	7.440E+18	1.310E+19	1.680E+19	1.810E+19	1.680E+19	1.310E+19	7.440E+18	8.620E+17
	TUD-S ₂	6.630E+17	7.430E+18	1.310E+19	1.680E+19	1.810E+19	1.680E+19	1.310E+19	7.430E+18	6.630E+17
	TUD-S ₆	6.830E+17	7.460E+18	1.300E+19	1.670E+19	1.800E+19	1.670E+19	1.300E+19	7.460E+18	6.830E+17

Table A.13

Step 0.3 – Temperature distribution along centerlines AA' and BB'.

Observable	Code	Results along AA' (points coordinates are expressed in m)								
		(0,0,1)	(0,25,1)	(0,5,1)	(0,75,1)	(1,0,1)	(1,25,1)	(1,5,1)	(1,75,1)	(2,0,1)
T (K)	DTU-MC	9.363E+02	1.192E+03	1.354E+03	1.362E+03	1.306E+03	1.225E+03	1.132E+03	1.034E+03	9.345E+02
	SEALION-mPK	9.236E+02	1.195E+03	1.355E+03	1.358E+03	1.302E+03	1.224E+03	1.132E+03	1.039E+03	9.264E+02
	CNRS-SP ₁	9.253E+02	1.194E+03	1.358E+03	1.363E+03	1.305E+03	1.224E+03	1.131E+03	1.034E+03	9.251E+02
	CNRS-SP ₃	9.236E+02	1.194E+03	1.357E+03	1.361E+03	1.304E+03	1.224E+03	1.131E+03	1.034E+03	9.235E+02
	PoliMi-MGD	9.253E+02	1.196E+03	1.361E+03	1.364E+03	1.305E+03	1.224E+03	1.132E+03	1.035E+03	9.252E+02
	PSI-MGD	9.253E+02	1.196E+03	1.356E+03	1.363E+03	1.306E+03	1.226E+03	1.133E+03	1.037E+03	9.252E+02
	TUD-S ₂	9.212E+02	1.194E+03	1.359E+03	1.364E+03	1.305E+03	1.224E+03	1.131E+03	1.032E+03	9.225E+02
	TUD-S ₆	9.219E+02	1.194E+03	1.356E+03	1.360E+03	1.303E+03	1.223E+03	1.131E+03	1.034E+03	9.233E+02
Observable	Code	Results along BB' (points coordinates are expressed in m)								
		(1,0,0)	(1,0,25)	(1,0,5)	(1,0,75)	(1,1,0)	(1,1,25)	(1,1,5)	(1,1,75)	(1,2,0)
T (K)	DTU-MC	9.318E+02	1.139E+03	1.271E+03	1.305E+03	1.306E+03	1.311E+03	1.316E+03	1.260E+03	9.810E+02
	SEALION-mPK	9.263E+02	1.139E+03	1.271E+03	1.303E+03	1.302E+03	1.307E+03	1.319E+03	1.263E+03	9.527E+02
	CNRS-SP ₁	9.252E+02	1.139E+03	1.273E+03	1.305E+03	1.305E+03	1.314E+03	1.321E+03	1.265E+03	9.322E+02
	CNRS-SP ₃	9.236E+02	1.140E+03	1.272E+03	1.304E+03	1.304E+03	1.313E+03	1.320E+03	1.265E+03	9.322E+02
	PoliMi-MGD	9.253E+02	1.140E+03	1.275E+03	1.307E+03	1.305E+03	1.313E+03	1.321E+03	1.265E+03	9.303E+02
	PSI-MGD	9.252E+02	1.139E+03	1.273E+03	1.307E+03	1.306E+03	1.312E+03	1.319E+03	1.263E+03	9.481E+02
	TUD-S ₂	9.215E+02	1.139E+03	1.273E+03	1.306E+03	1.305E+03	1.315E+03	1.322E+03	1.265E+03	9.374E+02
	TUD-S ₆	9.222E+02	1.140E+03	1.272E+03	1.303E+03	1.303E+03	1.312E+03	1.319E+03	1.264E+03	9.390E+02

Table A.14

Step 1.1 – Delayed neutron source along AA' and BB.

Observable	Code	Results along AA' (points coordinates are expressed in m)								
		(0,0,1)	(0,25,1)	(0,5,1)	(0,75,1)	(1,0,1)	(1,25,1)	(1,5,1)	(1,75,1)	(2,0,1)
$\sum_i \lambda_i C_i$ ($m^{-3}s^{-1}$)	DTU-MC	1.69E+016	1.430E+17	2.200E+17	2.420E+17	2.270E+17	1.920E+17	1.430E+17	8.060E+16	1.22E+016
	SEALION-mPK	1.649E+16	1.440E+17	2.204E+17	2.412E+17	2.268E+17	1.928E+17	1.468E+17	9.472E+16	1.949E+16
	CNRS-SP ₁	1.335E+16	1.452E+17	2.212E+17	2.411E+17	2.268E+17	1.923E+17	1.461E+17	9.214E+16	1.316E+16
	CNRS-SP ₃	1.251E+16	1.454E+17	2.209E+17	2.406E+17	2.264E+17	1.921E+17	1.462E+17	9.245E+16	1.233E+16
	PoliMi-MGD	1.321E+16	1.450E+17	2.219E+17	2.414E+17	2.266E+17	1.920E+17	1.459E+17	9.188E+16	1.292E+16
	PSI-MGD	1.325E+16	1.453E+17	2.214E+17	2.413E+17	2.270E+17	1.925E+17	1.463E+17	9.218E+16	1.314E+16
	TUD-S ₂	1.093E+16	1.438E+17	2.228E+17	2.426E+17	2.278E+17	1.927E+17	1.464E+17	8.968E+16	1.184E+16
	TUD-S ₆	1.132E+16	1.437E+17	2.212E+17	2.405E+17	2.261E+17	1.916E+17	1.461E+17	9.029E+16	1.224E+16
Observable	Code	Results along BB' (points coordinates are expressed in m)								
		(1,0,0)	(1,0,25)	(1,0,5)	(1,0,75)	(1,1,0)	(1,1,25)	(1,1,5)	(1,1,75)	(1,2,0)
$\sum_i \lambda_i C_i$ ($m^{-3}s^{-1}$)	DTU-MC	1.55E+016	1.150E+17	1.840E+17	2.180E+17	2.270E+17	2.250E+17	2.150E+17	1.740E+17	1.79E+016
	SEALION-mPK	2.081E+16	1.231E+17	1.915E+17	2.213E+17	2.268E+17	2.252E+17	2.145E+17	1.743E+17	4.482E+16
	CNRS-SP ₁	1.306E+16	1.190E+17	1.881E+17	2.193E+17	2.268E+17	2.261E+17	2.178E+17	1.754E+17	3.079E+16
	CNRS-SP ₃	1.222E+16	1.193E+17	1.879E+17	2.189E+17	2.264E+17	2.257E+17	2.175E+17	1.753E+17	3.072E+16
	PoliMi-MGD	1.297E+16	1.186E+17	1.881E+17	2.194E+17	2.266E+17	2.260E+17	2.177E+17	1.756E+17	2.805E+16
	PSI-MGD	1.299E+16	1.189E+17	1.881E+17	2.195E+17	2.270E+17	2.261E+17	2.176E+17	1.752E+17	2.730E+16
	TUD-S ₂	1.109E+16	1.174E+17	1.882E+17	2.203E+17	2.278E+17	2.281E+17	2.193E+17	1.768E+17	2.655E+16
	TUD-S ₆	1.143E+16	1.178E+17	1.872E+17	2.186E+17	2.261E+17	2.264E+17	2.179E+17	1.761E+17	2.728E+16

Table A.15

Step 1.2 – Profiles along AA' and BB' of the temperature and the change of fission rate density with respect to the solution obtained at Step 0.2.

Observable	Code	Results along AA' (points coordinates are expressed in m)								
		(0,0,1)	(0,25,1)	(0,5,1)	(0,75,1)	(1,0,1)	(1,25,1)	(1,5,1)	(1,75,1)	(2,0,1)
T (K)	DTU-MC	9.401E+02	1.193E+03	1.338E+03	1.349E+03	1.301E+03	1.227E+03	1.139E+03	1.044E+03	9.387E+02
	SEALION-mPK	9.280E+02	1.190E+03	1.346E+03	1.356E+03	1.300E+03	1.221E+03	1.131E+03	1.043E+03	9.278E+02
	CNRS-SP ₁	9.280E+02	1.195E+03	1.341E+03	1.349E+03	1.298E+03	1.225E+03	1.136E+03	1.041E+03	9.278E+02
	CNRS-SP ₃	9.262E+02	1.195E+03	1.341E+03	1.348E+03	1.298E+03	1.225E+03	1.137E+03	1.042E+03	9.260E+02
	PoliMi-MGD	9.281E+02	1.198E+03	1.343E+03	1.350E+03	1.300E+03	1.226E+03	1.138E+03	1.045E+03	9.280E+02
	PSI-MGD	9.282E+02	1.197E+03	1.340E+03	1.349E+03	1.300E+03	1.227E+03	1.139E+03	1.045E+03	9.280E+02
	TUD-S ₂	9.235E+02	1.196E+03	1.343E+03	1.350E+03	1.300E+03	1.226E+03	1.137E+03	1.041E+03	9.250E+02

(continued on next page)

Table A.15 (continued)

Observable	Code	Results along AA' (points coordinates are expressed in m)								
		(0,0,1)	(0,25,1)	(0,5,1)	(0,75,1)	(1,0,1)	(1,25,1)	(1,5,1)	(1,75,1)	(2,0,1)
$\Delta \int_E \Sigma_f \phi dE$ ($s^{-1}m^{-3}$)	TUD-S ₆	9.243E+02	1.196E+03	1.340E+03	1.347E+03	1.298E+03	1.225E+03	1.137E+03	1.042E+03	9.258E+02
	DTU-MC	4.630E+16	9.190E+16	-8.580E+17	-1.150E+18	-1.050E+18	-2.430E+17	4.560E+17	4.040E+17	5.930E+16
	SEALION-mPK	4.567E+16	1.503E+17	-5.552E+17	-9.958E+17	-7.411E+17	-5.098E+16	5.362E+17	6.960E+17	1.267E+17
	CNRS-SP ₁	7.800E+16	1.170E+17	-5.690E+17	-9.490E+17	-7.980E+17	-2.850E+16	2.630E+17	4.610E+17	7.780E+16
	CNRS-SP ₃	7.020E+16	1.160E+17	-5.670E+17	-9.430E+17	-7.940E+17	-2.870E+16	2.580E+17	4.580E+17	6.990E+16
	PoliMi-MGD	7.560E+16	1.140E+17	-5.680E+17	-9.440E+17	-7.910E+17	-2.800E+16	2.630E+17	4.560E+17	7.520E+16
	PSI-MGD	2.188E+15	1.230E+17	-5.490E+17	-9.270E+17	-7.810E+17	-2.720E+16	2.700E+17	4.630E+17	1.835E+15
	TUD-S ₂	7.200E+16	1.070E+17	-5.650E+17	-9.530E+17	-8.190E+17	-2.720E+16	2.630E+17	4.530E+17	7.190E+16
	TUD-S ₆	7.300E+16	1.040E+17	-5.570E+17	-9.380E+17	-8.100E+17	-2.790E+16	2.470E+17	4.430E+17	7.270E+16
Observable	Code	Results along BB' (points coordinates are expressed in m)								
		(1,0,0)	(1,0,25)	(1,0,5)	(1,0,75)	(1,1,0)	(1,1,25)	(1,1,5)	(1,1,75)	(1,2,0)
T (K)	DTU-MC	9.362E+02	1.152E+03	1.275E+03	1.305E+03	1.301E+03	1.302E+03	1.301E+03	1.245E+03	9.834E+02
	SEALION-mPK	9.281E+02	1.139E+03	1.268E+03	1.301E+03	1.300E+03	1.305E+03	1.310E+03	1.252E+03	9.350E+02
	CNRS-SP ₁	9.281E+02	1.148E+03	1.272E+03	1.301E+03	1.298E+03	1.304E+03	1.307E+03	1.253E+03	9.350E+02
	CNRS-SP ₃	9.262E+02	1.149E+03	1.272E+03	1.300E+03	1.298E+03	1.303E+03	1.306E+03	1.253E+03	9.351E+02
	PoliMi-MGD	9.281E+02	1.150E+03	1.275E+03	1.304E+03	1.300E+03	1.304E+03	1.307E+03	1.253E+03	9.470E+02
	PSI-MGD	9.282E+02	1.148E+03	1.273E+03	1.303E+03	1.300E+03	1.303E+03	1.306E+03	1.252E+03	9.517E+02
	TUD-S ₂	9.240E+02	1.148E+03	1.274E+03	1.302E+03	1.300E+03	1.306E+03	1.309E+03	1.254E+03	9.424E+02
	TUD-S ₆	9.247E+02	1.149E+03	1.272E+03	1.300E+03	1.298E+03	1.303E+03	1.306E+03	1.253E+03	9.442E+02
$\Delta \int_E \Sigma_f \phi dE$ ($s^{-1}m^{-3}$)	DTU-MC	1.730E+14	2.710E+17	-4.730E+17	-4.280E+17	-1.110E+18	-1.150E+18	-4.500E+17	-6.450E+16	6.730E+16
	SEALION-mPK	1.185E+17	3.795E+17	-7.906E+17	-5.642E+17	-6.375E+17	-6.511E+17	-5.618E+17	-8.285E+16	6.022E+16
	CNRS-SP ₁	8.200E+16	2.640E+17	-2.260E+17	-6.430E+17	-7.980E+17	-7.560E+17	-4.920E+17	-3.950E+16	8.350E+16
	CNRS-SP ₃	7.380E+16	2.620E+17	-2.270E+17	-6.410E+17	-7.940E+17	-7.520E+17	-4.900E+17	-3.920E+16	7.490E+16
	PoliMi-MGD	7.960E+16	2.640E+17	-2.260E+17	-6.390E+17	-7.940E+17	-7.540E+17	-4.920E+17	-4.160E+16	8.050E+16
	PSI-MGD	9.170E+16	2.640E+17	-2.290E+17	-6.510E+17	-8.020E+17	-7.480E+17	-4.860E+17	-3.480E+16	9.060E+16
	TUD-S ₂	7.590E+16	2.510E+17	-2.340E+17	-6.590E+17	-8.190E+17	-7.700E+17	-4.960E+17	-4.660E+16	6.930E+16
	TUD-S ₆	7.680E+16	2.440E+17	-2.370E+17	-6.540E+17	-8.100E+17	-7.610E+17	-4.910E+17	-4.530E+16	6.990E+16

Table A.16

Step 1.3 – Velocity components, temperature distribution, and delayed neutron source along AA'.

Observable	Code	Results along AA' (points coordinates are expressed in m)								
		(0,0,1)	(0,25,1)	(0,5,1)	(0,75,1)	(1,0,1)	(1,25,1)	(1,5,1)	(1,75,1)	(2,0,1)
u_x (ms^{-1})	DTU-MC	0.000E+00	1.558E-02	2.162E-02	1.472E-02	2.666E-04	-1.476E-02	-2.189E-02	-1.563E-02	0.000E+00
	SEALION-mPK	9.781E-06	1.785E-02	2.497E-02	1.667E-02	2.912E-04	-1.616E-02	-2.468E-02	-1.776E-02	-9.938E-06
	CNRS-SP ₁	0.000E+00	1.641E-02	2.310E-02	1.579E-02	1.250E-09	-1.579E-02	-2.310E-02	-1.641E-02	0.000E+00
	CNRS-SP ₃	0.000E+00	1.636E-02	2.302E-02	1.572E-02	1.000E-09	-1.572E-02	-2.302E-02	-1.636E-02	0.000E+00
	PoliMi-MGD	0.000E+00	1.637E-02	2.312E-02	1.578E-02	-1.130E-10	-1.578E-02	-2.312E-02	-1.637E-02	0.000E+00

Table A.16 (continued)

Observable	Code	Results along AA' (points coordinates are expressed in m)								
		(0,0,1)	(0,25,1)	(0,5,1)	(0,75,1)	(1,0,1)	(1,25,1)	(1,5,1)	(1,75,1)	(2,0,1)
u_y (ms ⁻¹)	PSI-MGD	0.000E+00	1.630E-02	2.263E-02	1.519E-02	-8.530E-09	-1.519E-02	-2.263E-02	-1.630E-02	0.000E+00
	TUD-S ₂	6.050E-06	1.644E-02	2.316E-02	1.584E-02	-2.220E-06	-1.584E-02	-2.316E-02	-1.644E-02	-6.050E-06
	TUD-S ₆	5.980E-06	1.631E-02	2.295E-02	1.566E-02	-2.200E-06	-1.566E-02	-2.295E-02	-1.631E-02	-5.980E-06
	DTU-MC	0.000E+00	-1.772E-01	1.972E-02	1.418E-01	1.721E-01	1.408E-01	1.908E-02	-1.789E-01	0.000E+00
	SEALION-mPK	-1.409E-02	-1.773E-01	1.733E-02	1.402E-01	1.687E-01	1.402E-01	1.786E-02	-1.766E-01	-1.408E-02
	CNRS-SP ₁	0.000E+00	-1.777E-01	1.721E-02	1.376E-01	1.649E-01	1.376E-01	1.721E-02	-1.777E-01	0.000E+00
	CNRS-SP ₃	0.000E+00	-1.771E-01	1.708E-02	1.372E-01	1.645E-01	1.372E-01	1.708E-02	-1.771E-01	0.000E+00
	PoliMi-MGD	0.000E+00	-1.767E-01	1.741E-02	1.368E-01	1.638E-01	1.368E-01	1.741E-02	-1.767E-01	0.000E+00
	PSI-MGD	0.000E+00	-1.779E-01	1.662E-02	1.376E-01	1.659E-01	1.376E-01	1.662E-02	-1.779E-01	0.000E+00
	TUD-S ₂	-2.890E-05	-1.780E-01	1.735E-02	1.379E-01	1.650E-01	1.379E-01	1.735E-02	-1.780E-01	-2.890E-05
	TUD-S ₆	-2.930E-05	-1.766E-01	1.694E-02	1.368E-01	1.639E-01	1.368E-01	1.694E-02	-1.766E-01	-2.930E-05
T (K)	DTU-MC	9.354E+02	1.190E+03	1.278E+03	1.283E+03	1.283E+03	1.284E+03	1.282E+03	1.189E+03	9.351E+02
	SEALION-mPK	9.279E+02	1.186E+03	1.274E+03	1.281E+03	1.276E+03	1.281E+03	1.274E+03	1.186E+03	9.279E+02
	CNRS-SP ₁	9.279E+02	1.193E+03	1.278E+03	1.284E+03	1.280E+03	1.284E+03	1.278E+03	1.193E+03	9.279E+02
	CNRS-SP ₃	9.260E+02	1.193E+03	1.278E+03	1.284E+03	1.280E+03	1.284E+03	1.278E+03	1.193E+03	9.260E+02
	PoliMi-MGD	9.279E+02	1.193E+03	1.279E+03	1.286E+03	1.282E+03	1.286E+03	1.279E+03	1.193E+03	9.279E+02
	PSI-MGD	9.279E+02	1.191E+03	1.278E+03	1.284E+03	1.280E+03	1.284E+03	1.278E+03	1.191E+03	9.279E+02
	TUD-S ₂	9.248E+02	1.193E+03	1.279E+03	1.285E+03	1.281E+03	1.285E+03	1.279E+03	1.193E+03	9.248E+02
	TUD-S ₆	9.257E+02	1.192E+03	1.277E+03	1.283E+03	1.280E+03	1.283E+03	1.277E+03	1.192E+03	9.257E+02
	DTU-MC	0.000E+00	1.493E+17	2.026E+17	2.098E+17	1.913E+17	2.100E+17	2.017E+17	1.489E+17	0.000E+00
	SEALION-mPK	1.440E+16	1.460E+17	2.000E+17	2.100E+17	1.920E+17	2.100E+17	2.000E+17	1.460E+17	1.440E+16
$\sum_i \lambda_i C_i$ (m ⁻³ s ⁻¹)	CNRS-SP ₁	1.500E+16	1.470E+17	2.000E+17	2.120E+17	1.990E+17	2.120E+17	2.000E+17	1.470E+17	1.500E+16
	CNRS-SP ₃	1.410E+16	1.470E+17	2.000E+17	2.120E+17	1.990E+17	2.120E+17	2.000E+17	1.470E+17	1.410E+16
	PoliMi-MGD	1.440E+16	1.460E+17	1.990E+17	2.110E+17	1.940E+17	2.110E+17	1.990E+17	1.460E+17	1.440E+16
	PSI-MGD	1.480E+16	1.470E+17	2.000E+17	2.130E+17	1.980E+17	2.130E+17	2.000E+17	1.470E+17	1.480E+16
	TUD-S ₂	1.040E+16	1.490E+17	2.020E+17	2.140E+17	1.940E+17	2.140E+17	2.020E+17	1.490E+17	1.040E+16
	TUD-S ₆	1.090E+16	1.480E+17	2.010E+17	2.130E+17	1.930E+17	2.130E+17	2.010E+17	1.480E+17	1.090E+16
Observable	Code	Results along BB' (points coordinates are expressed in m)								
		(1,0,0)	(1,0,25)	(1,0,5)	(1,0,75)	(1,1,0)	(1,1,25)	(1,1,5)	(1,1,75)	(1,2,0)
u_y (ms ⁻¹)	DTU-MC	0.000E+00	3.690E-02	9.397E-02	1.422E-01	1.721E-01	1.738E-01	1.368E-01	6.071E-02	0.000E+00
	SEALION-mPK	1.119E-05	3.478E-02	8.972E-02	1.377E-01	1.687E-01	1.723E-01	1.368E-01	6.075E-02	1.909E-05
	CNRS-SP ₁	0.000E+00	3.512E-02	8.947E-02	1.359E-01	1.649E-01	1.665E-01	1.307E-01	5.756E-02	0.000E+00
	CNRS-SP ₃	0.000E+00	3.510E-02	8.933E-02	1.356E-01	1.645E-01	1.660E-01	1.303E-01	5.740E-02	0.000E+00
	PoliMi-MGD	0.000E+00	3.507E-02	8.909E-02	1.351E-01	1.638E-01	1.656E-01	1.302E-01	5.743E-02	0.000E+00
	PSI-MGD	0.000E+00	3.537E-02	9.055E-02	1.374E-01	1.659E-01	1.669E-01	1.309E-01	5.780E-02	0.000E+00
	TUD-S ₂	-1.230E-05	3.510E-02	8.950E-02	1.360E-01	1.650E-01	1.667E-01	1.308E-01	5.763E-02	-2.810E-05
	TUD-S ₆	-1.330E-05	3.506E-02	8.912E-02	1.352E-01	1.639E-01	1.655E-01	1.299E-01	5.719E-02	-2.870E-05
	DTU-MC	9.352E+02	1.068E+03	1.158E+03	1.229E+03	1.283E+03	1.316E+03	1.322E+03	1.276E+03	9.573E+02
	SEALION-mPK	9.280E+02	1.063E+03	1.148E+03	1.219E+03	1.276E+03	1.316E+03	1.328E+03	1.285E+03	9.284E+02
T(K)	CNRS-SP ₁	9.280E+02	1.067E+03	1.156E+03	1.226E+03	1.280E+03	1.315E+03	1.326E+03	1.283E+03	9.284E+02
	CNRS-SP ₃	9.261E+02	1.067E+03	1.156E+03	1.226E+03	1.280E+03	1.315E+03	1.325E+03	1.282E+03	9.266E+02
	PoliMi-MGD	9.280E+02	1.067E+03	1.157E+03	1.228E+03	1.282E+03	1.317E+03	1.327E+03	1.284E+03	9.282E+02
	PSI-MGD	9.281E+02	1.068E+03	1.156E+03	1.226E+03	1.280E+03	1.314E+03	1.324E+03	1.281E+03	9.287E+02
	TUD-S ₂	9.250E+02	1.066E+03	1.156E+03	1.227E+03	1.281E+03	1.316E+03	1.327E+03	1.283E+03	9.137E+02
	TUD-S ₆	9.258E+02	1.069E+03	1.157E+03	1.226E+03	1.280E+03	1.314E+03	1.325E+03	1.282E+03	9.149E+02
	DTU-MC	0.000E+00	7.838E+16	1.258E+17	1.639E+17	1.913E+17	2.052E+17	2.020E+17	1.683E+17	0.000E+00
	SEALION-mPK	1.050E+16	8.020E+16	1.280E+17	1.660E+17	1.920E+17	2.050E+17	2.000E+17	1.650E+17	1.580E+16
	CNRS-SP ₁	1.480E+16	8.800E+16	1.380E+17	1.750E+17	1.990E+17	2.090E+17	2.020E+17	1.660E+17	1.720E+16
	CNRS-SP ₃	1.380E+16	8.840E+16	1.380E+17	1.750E+17	1.990E+17	2.080E+17	2.010E+17	1.660E+17	1.630E+16
$\sum_i \lambda_i C_i$ (m ⁻³ s ⁻¹)	PoliMi-MGD	1.450E+16	8.620E+16	1.340E+17	1.710E+17	1.940E+17	2.050E+17	1.980E+17	1.620E+17	1.580E+16
	PSI-MGD	1.480E+16	8.780E+16	1.370E+17	1.740E+17	1.980E+17	2.080E+17	2.010E+17	1.660E+17	1.690E+16
	TUD-S ₂	1.340E+16	8.510E+16	1.330E+17	1.700E+17	1.940E+17	2.050E+17	1.980E+17	1.620E+17	1.240E+16
	TUD-S ₆	1.380E+16	8.600E+16	1.340E+17	1.690E+17	1.930E+17	2.030E+17	1.970E+17	1.620E+17	1.300E+16

The left plot in Fig. 10 shows the temperature profile along the horizontal centerline. A good agreement with the benchmark results is noted, with DTU and SEALION results yielding a code-wise discrepancy below 0.4% for both codes, which is comparable to the average relative discrepancy below 0.1% reported by the

benchmark participants. The right plot in Fig. 10 shows the fission rate density change compared to the standalone neutronics test in Step 0.2 with static fuel. Due to the nature of the Monte-Carlo based neutronics approach in the DTU code it is expected that the change in fission rate density, which is more than an order of

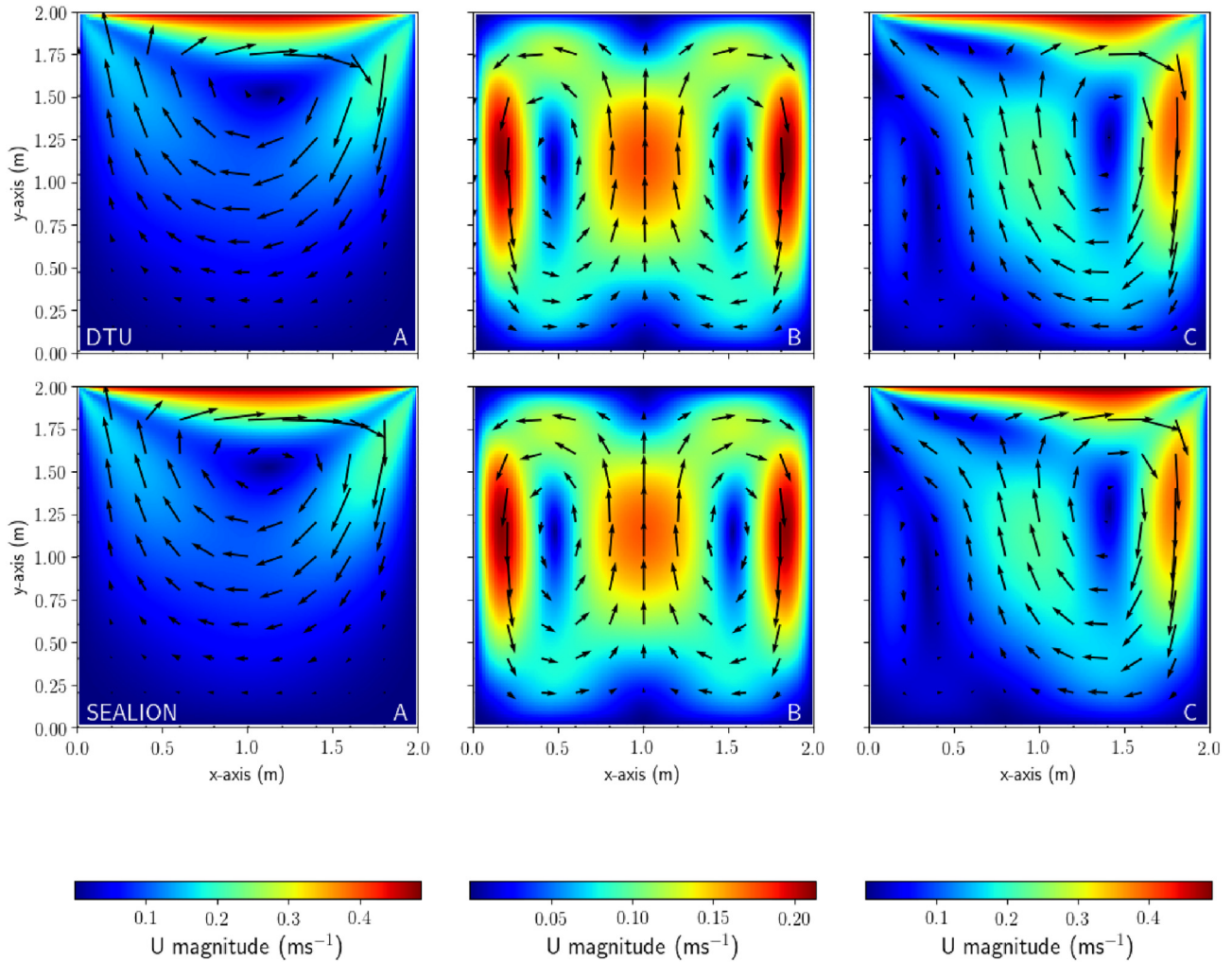


Fig. B.13. Velocity magnitude field and flow streamlines obtained by each code in three different benchmark steps. From left to right: Step 0.1(A); Step 1.3(B); and Step 1.4(C) with $P = 1 \text{ GW}$ and $U_{hd} = 0.5 \text{ ms}^{-1}$. DTU results are shown in the top row and results from SEALION are shown in the bottom row.

magnitude smaller than the fission rate density itself, is to have large statistical fluctuations. The same is true for the SEALION code which cannot solve for the change in fission rate density directly but infers the results from an additional Monte-Carlo simulation. Error-bands representing the two standard deviation limit have been imposed on the figure to illustrate the magnitude of the statistical uncertainties. The DTU/SEALION results and the benchmark results agree within the statistical uncertainties of our models. Note that the DTU/SEALION curves shown in the figure are derived as a cubic spline average of the stochastic results, hence the curves themselves do not signify a difference in the shape of the fission rate density. The code-wise discrepancy are in both cases larger than the average discrepancy of 1%-2% reported by the benchmark participants. The overall shape of the change in fission rate is largely reproduced by both the DTU and the SEALION code. This means that the feedback mechanism is captured correctly; increasing the temperature causes the power to decrease via a negative density feedback coefficient. This claim is supported by the results from Sections 5.1.3 and 5.2.1 where it was demonstrated that both the temperature distribution and delayed neutron source agree with the benchmark results.

Computing the change in fission rate density is a stress test of the codes; and testing the capability of the codes to model higher order effects, which are more than a magnitude smaller than the fission rate density itself. The SEALION code does not take small

changes to the fission cross section due to local temperature changes, into account. The SEALION code therefore assumes that the fundamental form does not change during a simulation and the code is as such not expected to reproduce higher order effects. In the case of the DTU code the quality of the results are limited by a compromise between computational resources and the desired precision. An intermediate conclusion would be that both the DTU and the SEALION code falls short in describing the higher order perturbation to the fission rate density caused by the shift of DNPs from a region of high neutron importance in the center of the cavity to the region of lower neutron importance away from the center. It is thus expected that higher order deterministic codes perform better in describing high precision results such as the change in fission rate density. The effect of the higher order corrections and thereby the validity of the point kinetics approximation applied in the SEALION framework can be gauged by comparing the results from the SEALION code, to the higher order codes used by the benchmark participants in Fig. 10. A detailed analysis of this effect, however, is outside the scope of the present paper. Table A.15 in Appendix A shows the temperature and change in fission rate density for the various codes.

Table 8 gives the reactivity changes compared to the first step of the steady-state phase. The results from the DTU and SEALION codes are compatible with the results from the benchmark participants within 2–3 standard deviations.

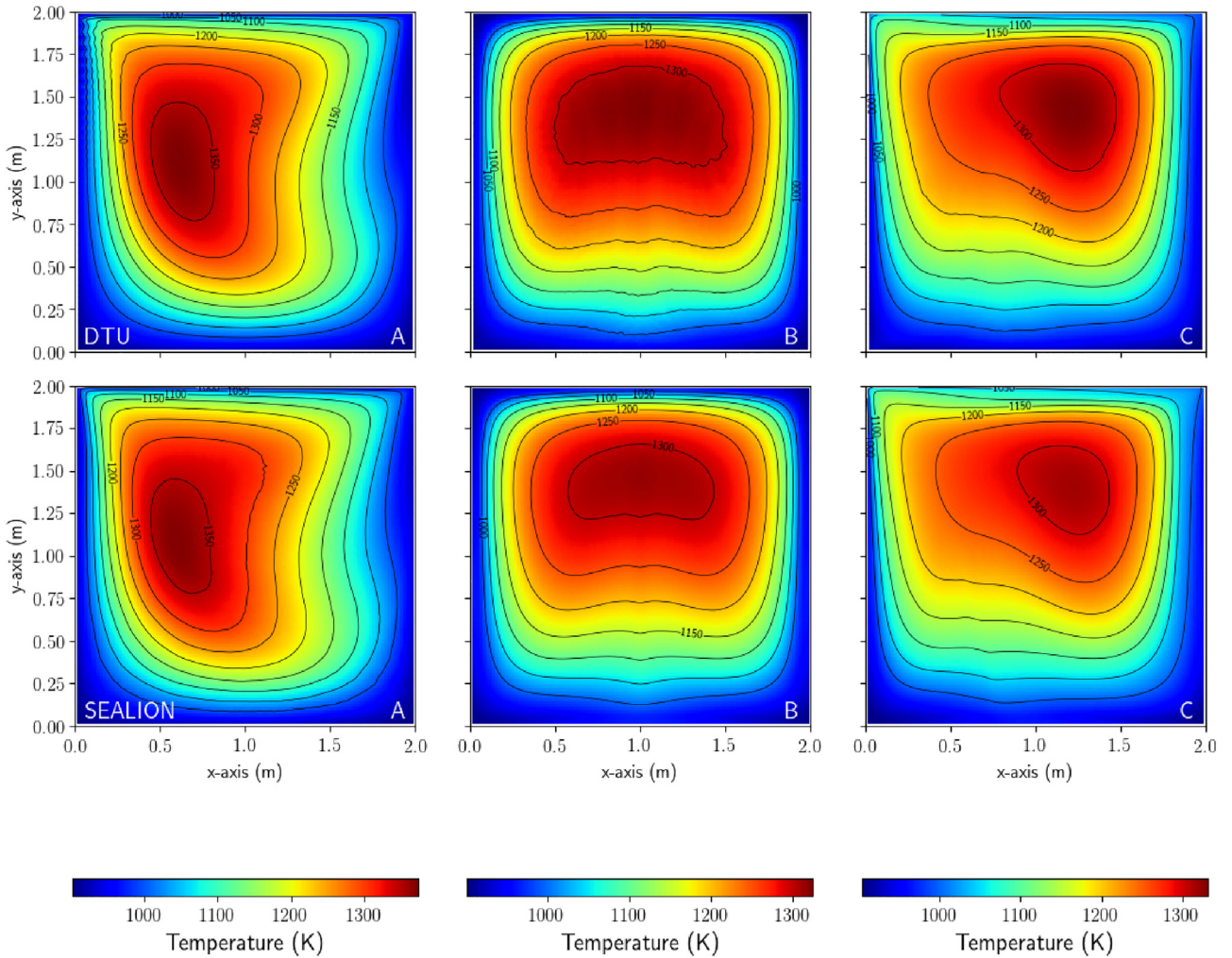


Fig. B.14. Temperature field with isolines obtained by each code in three different benchmark steps. From left to right: Step 0.3(A); Step 1.3(B); and Step 1.4(C) with $P = 1$ GW and $U_{lid} = 0.5$ ms⁻¹. DTU results are shown in the top row and results from SEALION are shown in the bottom row.

All values used in plots for this step can be found in [Table A.12](#) in [Appendix A](#).

5.2.3. Step 1.3: Buoyancy

Buoyancy effects are considered, and the velocity and temperature fields are assessed along with the reactivity changes due to DNP transport under the assumption of natural convection only. In [Fig. 11](#) the vertical component of the velocity and the temperature are shown. In terms of velocity, the agreement is fair yielding a code-wise discrepancy of $\epsilon_c(DTU) \in [2.5\%, 5.2\%]$ along the AA' centerline and $\epsilon_c(DTU) = 4.2\%$ along the BB' centerline. For the SEALION code the corresponding numbers are $\epsilon_c(SEALION) \in [2.1\%, 5.3\%]$ and $\epsilon_c(SEALION) = 3.4\%$. These numbers are to be compared to a code-wise discrepancy for benchmark participants of 0.2%–1.8% and an average relative discrepancy below 0.6%. The x-component of the velocity field along the BB' centerline is omitted from the benchmark due to the symmetry of the solution as can be seen from [Fig. B.13](#). In [Fig. 11](#) also the temperature profile along the horizontal centerline is shown. Here the agreement between the results is very good with code-wise discrepancies below 0.4% for both codes. The slight overshoot of the DTU code of u_y along the horizontal centerline can be explained by the slight overshoot of the temperature profile along the same centerline. In [Fig. 11](#), the delayed neutron precursor source is shown along the horizontal centerline. The agreement between

the participants is good; the benchmark participants report 0.5 % discrepancies between the results, DTU results show some deviation close to the center which can be attributed to a non-converged solution. In this step the SEALION results agree slightly better with the benchmark participants yielding code-wise discrepancies of no larger than a few percent.

[Table 9](#) reports the change in reactivity compared to the standalone neutronics test. There is around 50 pcm of difference reported between the benchmark participants. The estimated reactivity loss due to buoyancy effects for both the DTU and SEALION codes are statistically compatible with the results from the benchmark participants.

All temperature, velocity and delayed neutron source values from this step are shown in [Table A.16](#) in [Appendix A](#).

5.2.4. Step 1.4: Full coupling

The objective of this step is to assess the capabilities of the codes to predict the loss of reactivity in a setting that constitutes a realistic reactor simulation under different operating conditions. The different operating conditions are emulated at this stage by increasing the velocity of the lid U_{lid} from 0 to 1 ms⁻¹ in steps of 0.1 ms⁻¹ and the power from 0 to 1 GW with steps of 0.2 GW. The impact on the system reactivity is investigated. The change in reactivity as a function of power is provided in [Table 10](#) for cavity lid velocities $U_{lid} = 0$ ms⁻¹, 0.3 ms⁻¹ and $U_{lid} = 0.5$ ms⁻¹.

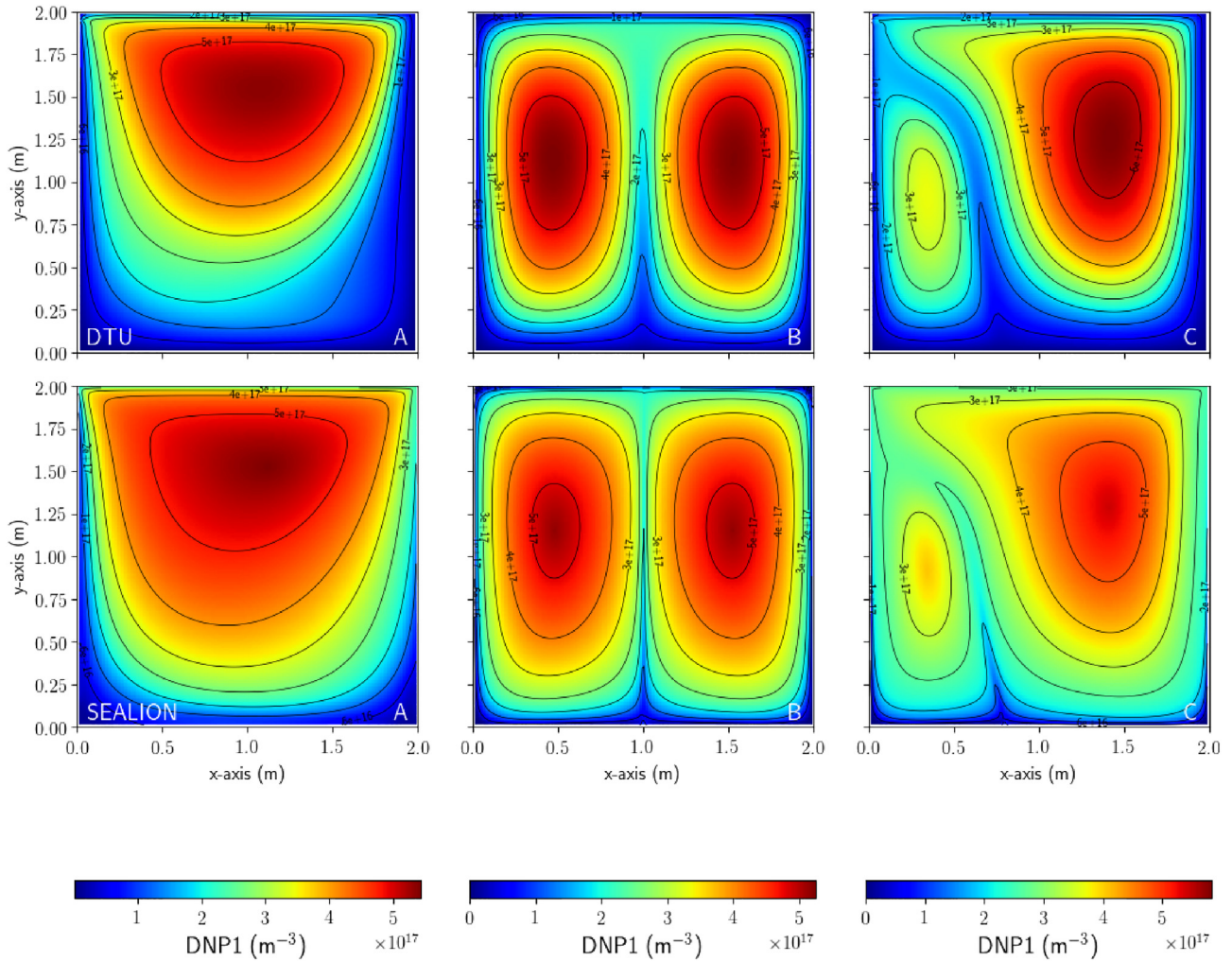


Fig. B.15. Distribution of the first family of delayed neutron precursors ($T_{1/2} = 55.6$ s) with isolines obtained by each code in three different benchmark steps. From left to right: Step 1.1(A); Step 1.3(B); and Step 1.4(C) with $P = 1$ GW and $U_{lid} = 0.5$ ms $^{-1}$. DTU results are shown in the top row and results from SEALION are shown in the bottom row.

When inspecting the reactivity differences $\Delta\rho_{s_{0.2}} = \rho - \rho_{s_{0.2}}$ in Table 10, two interesting phenomena appear. In general, an increase in power level leads to an increase in the temperature of the system, causing fuel salt expansion and increased buoyancy. The increased buoyancy effectively drives the DNP1s away from the cavity center onto regions of lower neutron importance, lowering the reactivity and leading to larger negative value of $\Delta\rho_{s_{0.2}}$ for higher power levels. The same effect in the reactivity caused by a shift of the DNP1s to regions of lower importance is observed at lower power levels (column 1) when the forced convection is increased by changing the speed of the cavity lid from 0 ms $^{-1}$ to 0.5 ms $^{-1}$. At higher power levels, starting from $P = 0.6$ GW, higher velocities result in slightly less negative reactivity. This is due to two competing processes: a high velocity shifts the DNP1s away from the cavity center, whereas a high power reduces the temperature in the regions of higher importance. Depending which effect is stronger, the negative reactivity level will increase or decrease (Tiberga et al., 2020).

In terms of reactivity, the code-wise discrepancies for the DTU code are $\epsilon_c(DTU) \in [7\%, 9\%]$ and $\epsilon_c(SEALION) \in [4\%, 8\%]$ for SEALION, cf. Table 10. These numbers are to be compared with an average relative discrepancy for benchmark participants of 1%–13%. The discrepancy between the SEALION code and the benchmark participants, can be ascribed to the fact that the shape of the power

distribution is assumed to remain unchanged, thereby ignoring second order perturbations to the fundamental form. The relative importance of the second order effects to the fundamental form increases at lower power levels for higher levels of forced convection. This explains the slight difficulties of the SEALION code to model the increase in $|\Delta\rho_{s_{0.2}}|$ for increasing U_{lid} . The results from the DTU replicate the trend from the benchmark participants. This difference can be explained by the slight difference in velocity, fission density and delayed neutron source as discussed in Sections 5.2.1–5.2.3.

6. Phase 2: Time dependent coupling

The left panel in Fig. 12 shows the power gain as a function of frequency. The right panel in Fig. 12 compares the results for the power phase shift compared to the γ wave. A discrepancy for the SEALION code of $\epsilon_c(SEALION) = 4.6\%$ is observed for the gain diagrams. These numbers are to be compared to a code-wise and average relative discrepancy for benchmark participants of less than 1%. For the phase diagrams the agreement is better yielding a code-wise discrepancy for the SEALION code of $\epsilon_c(SEALION) = 8.2\%$. However, these numbers are to be compared to a code-wise discrepancy for the benchmark participants of 1.3%–3.2% and an average relative discrepancy below 2.2%.

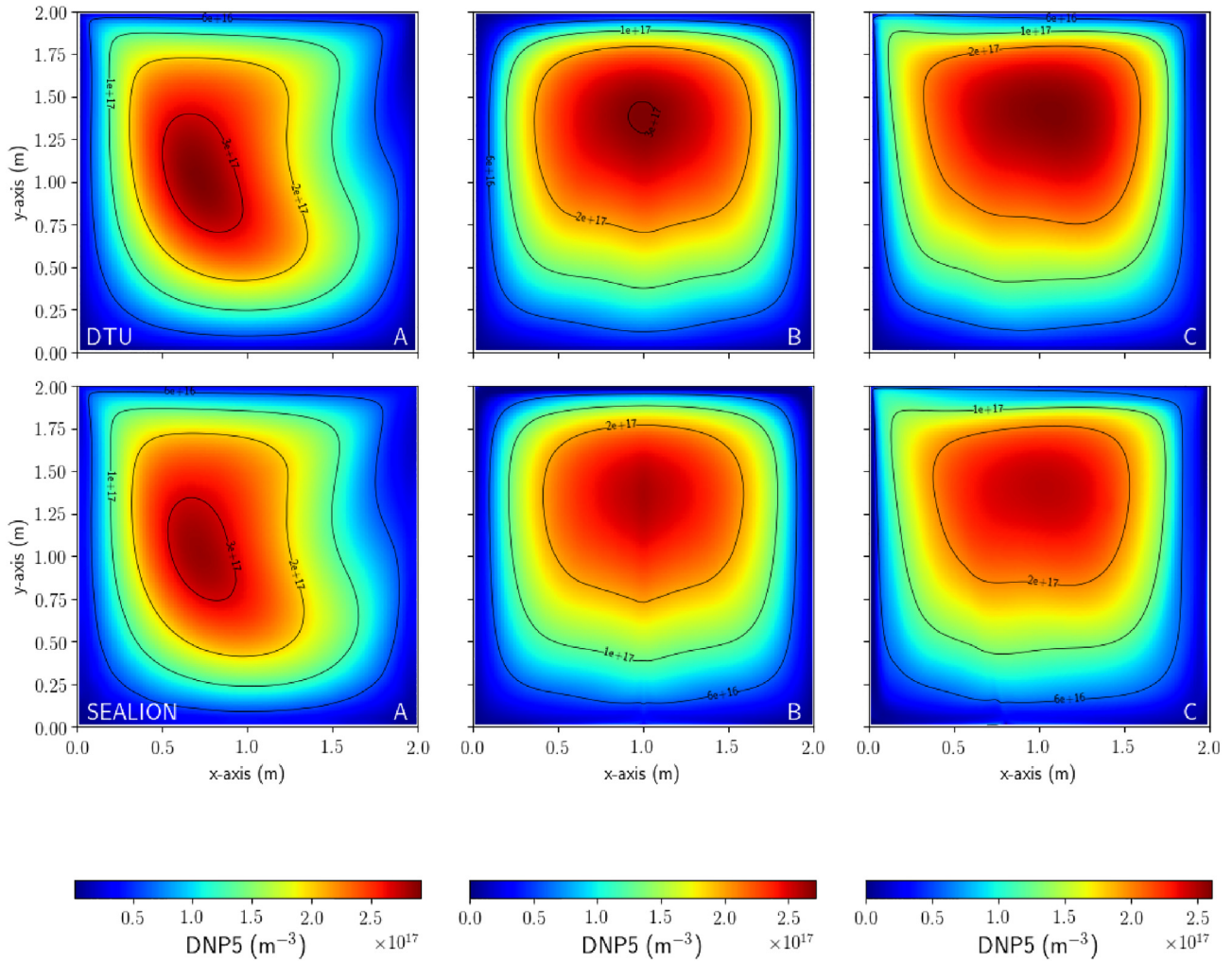


Fig. B.16. Distribution of the fifth family of delayed neutron precursors ($T_{1/2} = 2.37$ s) with isolines obtained by each code in three different benchmark steps. From left to right: Step 1.1(A); Step 1.3(B); and Step 1.4(C) with $P = 1$ GW and $U_{hd} = 0.5$ ms $^{-1}$ DTU results are shown in the top row and results from SEALION are shown in the bottom row.

As for DTU, due to the necessity to do very fine power change sampling and the computational costs related to that, at this stage only one frequency is compared; 0.025 Hz. For this frequency, the gain computed by coupled Serpent2-OpenFOAM approach is 0.713 and the phase shift is 27° as compared to the average of 0.835 and 18.6° obtained by the benchmark participants. This relatively large discrepancies are mainly attributed to the sampling time-step used in Serpent2-OpenFOAM coupling approach to extract the data points for the power change and is expected to become more pronounced at higher frequencies. However, it can be appreciated that the coupling technique is capable to capture reasonably well the physical phenomena related to the non-uniform salt cooling in a frequency range where the DNPs is at equilibrium at each instant. This is expected since the period for the slow transients are comparable to the half-lives of the longest lived DNP groups giving a gain of ~ 1 and small phase-shift. The SEALION results agrees fairly well with the benchmarks participants meaning that the SEALION code to a large extent is able to replicate the sinusoidal power trend induced by the negative density feedback coefficient from the slow transient to the high frequencies domain where the kinetics are governed by short lived DNPs. The SEALION code overshoots the phase shift slightly which can be explained by the fact that due to the assumption on the fundamental form in the modified point-kinetics solver the effects of DNPs movement and other sec-

ond order effects are underestimated in this approach. These effects are expected to only influence the shape of the power transfer function and not the reduction in the system response amplitude which can also be observed since the SEALION code describes the gain well over a large frequency range.

7. Conclusion

This paper presents the results of a code-to-code verification of two multiphysics coupling techniques for molten salt reactor modeling. The coupled Serpent2-OpenFOAM method and an in-house developed SEALION software are compared for several multiphysics software features in a benchmark defined for a generic fast MSR (Tiberga et al., 2020).

The standalone physics modules yield almost identical results with the benchmark participants. Gradual introduction of coupling terms shows that both coupling schemes well describe the neutronics feedback mechanisms. Both the DTU and the SEALION approaches show relatively small discrepancies, comparable to the codewise discrepancies between the benchmark participants. For the transients, a larger difference of up to 20% is observed for the higher frequencies; this is likely attributed to the limitations of the point kinetics approach used in the SEALION code. A detailed

analysis of this effect, however, is outside the scope of the present paper and will be postponed for future work.

The solvers capture most of the intrinsic features of the liquid fuel, *inter alia* the reactivity loss due to the circulation of the DNPs, the reactor feedback mechanisms and the distribution of the DNP fields. Given the simple benchmark geometry, the coupling schemes applied to the benchmark including the DTU and SEALION are able to model the physics of the liquid fueled reactor as long as any perturbation to the system is not both large and local. However, both the DTU and the SEALION code have limitations in describing the higher order effects in the fission rate density caused by a shift of DNPs from a region of high neutron importance in the center of the cavity to the region of lower neutron importance away from the center.

CRedit authorship contribution statement

J. Groth-Jensen: Software, Validation, Formal analysis, Investigation, Data curation, Visualization. **A. Nalbandyan:** Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing - original draft. **E.B. Klinkby:** Supervision, Writing - review & editing, Validation. **B. Lauritzen:** Supervision, Writing - review & editing, Project administration. **P. Sabbagh:** Conceptualization, Methodology, Software. **A.V. Pedersen:** Conceptualization, Methodology, Software.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Additional tables

In the Appendix A several tables are presented containing raw data for velocity, temperature and delayed neutron precursor source values by all participants for the various benchmark steps. See Tables A.11–A.16.

Appendix B. Observables fields

In the Appendix B several figures are presented containing raw data for velocity, temperature and delayed neutron precursor

source values by all participants for the various benchmark steps. See Figs. B.13–B.16

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